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What Could Policymakers Have Known? Employment Risk and Administrative Data in the Dutch NOW Programme

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Statement of Originality

During the preparation of this work the author used the following GenAI tools: Claude (Anthropic) in order to receive feedback on the clarity and structure of drafts written by the author, to identify possible language and style issues, to assist with writing and debugging the analysis code, to format the author's own computed results into tables , and to ask questions that helped the author refine and pressure-test their own ideas and arguments. All research ideas, methodological choices, analysis, computed results, and written text are the author's own. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication submission.

Abstract

Governments implementing emergency wage subsidies face severe information constraints when allocating support rapidly. This study develops a firm-level crisis risk framework and applies it to recipients of the Dutch NOW wage subsidy programme. Two logit models estimated on pre-crisis LISA data distinguish predicted structural from crisis-induced employment loss, which are combined into a measure of relative crisis job loss. This measure is used to analyse the distribution of Round 1 NOW expenditure and expected unemployment-benefit (WW) costs, while firms receiving NOW across all eight rounds are ranked by subsequent firm exit.

Round 1 NOW expenditure was distributed relatively evenly across the relative crisis job-loss ranking, but each euro became progressively more likely to reach crisis-induced employment loss and less likely to support employment not predicted to be lost. Expected WW costs followed a different pattern, with lower crisis tiers remaining associated with higher total expected WW exposure under a no-intervention scenario. Structural firm characteristics provided the strongest overall ranking of subsequent firm exits, while the contribution of reported revenue loss differed across sectors.

These findings show that administrative information available during programme implementation could distinguish structural from crisis-induced employment loss, identify where support was more likely to reach crisis-induced employment loss, relate this to one important channel of expected public costs, and distinguish firms that later exited from those that survived.

Introduction

The COVID-19 pandemic prompted governments across the world to deploy large-scale wage subsidy programmes at great speed. In the Netherlands, the Tijdelijke Noodmaatregel Overbrugging voor Werkgelegenheid (NOW) provided wage subsidies to firms reporting revenue losses, spending approximately €22 billion across eight rounds between 2020 and 2022 (UWV, 2022). Designed as a liquidity bridge, the programme prioritized rapid implementation over targeted support to firms most at risk of losing jobs (Staatscourant, 2020).

Speed came at the cost of precision. NOW paid any firm reporting at least a 20 percent revenue loss, a threshold the Algemene Rekenkamer (2022) notes was chosen for quick implementation. Broad eligibility means funds reached firms that would have contracted regardless of the pandemic alongside those whose employment losses were primarily crisis-induced. At the time, policymakers could not observe how much expenditure fell into each category. Whether this distinction could already be made from data available during programme implementation is therefore an empirical question.

Because emergency wage-subsidy programmes involve substantial public expenditure, improving the use of administrative information during programme implementation could make future subsidies better targeted, increasing the share of funding reaching crisis-induced employment loss while reducing unnecessary public expenditure. Across all corona support measures, Rabobank (2024) estimates at least 11,000 bankruptcies to be prevented from 2020 till 2027, but at high public cost and sustained too long, keeping non-viable firms active. Understanding how much can already be inferred from the data available in real time is therefore relevant for designing future emergency support that balances urgency with better targeting.

This study addresses three questions:

1. Can expected employment loss be separated into a structural component, predicted from pre-crisis firm characteristics, and a crisis-induced component, identified from reported revenue loss and a common crisis indicator?
2. How was Round 1 NOW expenditure distributed relative to predicted structural and crisis-induced employment loss, and what does this imply for expected unemployment-benefit (WW) costs?
3. Among firms that received NOW support, which information best separates those that subsequently exited from those that survived?

The contribution of this paper is to examine how much can be inferred from data available, before employment outcomes are observed. It investigates whether existing administrative data can distinguish structural from crisis-induced employment loss, relate these predictions to the allocation of NOW expenditure and expected unemployment-benefit (WW) costs, and find firms that later exited from those that survived.

2. Literature Review

2.1 Why governments pay firms to keep workers during a crisis

Wage subsidies respond to a specific market failure. In a sharp downturn, a viable firm can fail because a temporary drop in revenue leaves it unable to cover fixed labour costs (Gourinchas et al., 2021). When a firm closes, the value built up in its relationships with workers is lost: laid-off workers suffer lasting earnings and skill losses (Blanchard and Summers, 1986), and workers recalled by a former employer recover faster than those who must find a new job (Fujita and Moscarini, 2017). Wage subsidies keep these firms and jobs in place until demand returns.

The evidence suggests that broad, rapid support stabilised the economy during the starting phase of the crisis. Chetty et al. (2020) show that rapid US support stabilised spending and employment early in the pandemic, and Erken et al. (2024) estimate that Dutch support prevented at least 11,000 bankruptcies. Therefore broad support is a tradeoff: Verifying eligibility slows support when firms need support quickly (Gourinchas et al., 2021), and CPB (2020) projections at the time showed extreme uncertainty about the depth and duration of the downturn. NOW reflected this design, paying any firm reporting at least a 20 percent revenue loss, a threshold the Algemene Rekenkamer (2022) notes was chosen for speed rather than precision.

This literature establishes why broad wage subsidies were justified during the acute phase of the pandemic, but also why rapid implementation limits the precision with which support can be directed towards firms facing genuine crisis-related distress.

2.2 Structural weakness and crisis shocks as separate sources of risk

Because eligibility was broad, NOW reached two kinds of firm: some already structurally weak before the pandemic and others that were healthy beforehand but affected by the temporary shock. Caballero, Hoshi and Kashyap (2008) describe the first group as structurally weak firms operating in declining markets that would have contracted regardless of any crisis. Gourinchas et al. (2021) show that many firms affected by COVID-19 were viable before the pandemic and were distressed only because demand collapsed. Previous research therefore treats structural weakness and crisis shocks as different sources of exit risk. A broad subsidy, however, reaches both without distinguishing between them, so the share of spending directed towards genuine crisis distress rather than firms that would have contracted anyway cannot be observed directly.

Whether firms are kept alive matters for productivity, and the literature shows the effect can run in both directions. A study of Japan found that cheap credit kept unproductive firms alive and diverted customers and resources away from healthier competitors (Caballero, Hoshi and Kashyap, 2008). At the same time, González, Moral-Benito and Soler (2023) show that the relationship between firm exit and productivity is not constant: at normal exit rates, closures raise productivity, in line with Schumpeter's (1942) concept of creative destruction, whereas at very high exit rates the effect reverses as healthy firms are pulled down alongside failing ones. For the Netherlands, Freeman et al. (2021) show that exiting firms contribute to productivity growth and that Dutch business dynamism was already declining

before the pandemic, while Adema et al. (2021) suggest that NOW support may have prevented some closures that would otherwise have occurred.

The difference between structural weakness and crisis-induced distress therefore matters not only for employment protection but also for the longer-run economic consequences of broad crisis support.

2.3 Predicting firm exit from administrative data

Firm exit can be predicted from observable firm characteristics, and the literature identifies which variables contain predictive information. Altman (1968) showed that a small number of financial ratios can identify failing firms years in advance. Shumway (2001) improved on this by following firms over time, showing that firm age and size matter alongside financial variables. Lessmann et al. (2015) demonstrate that more complex machine-learning methods often improve predictive accuracy but at the cost of interpretability.

The most accurate models rely on financial information. Davies, Kattenberg and Vogt (2023) combine machine learning with administrative records and financial information to predict Dutch firm exits substantially better than standard approaches. They also show that much of this predictive performance can be retained using a considerably smaller set of predictors, suggesting that highly complex models are not always necessary. The same paper finds that financially weaker firms were less likely to apply for NOW, implying that programme recipients are not representative of the broader firm population.

Administrative data for this reason contains useful information about firm vulnerability, though existing models mostly rely on a combination of firm characteristics and financial information. Existing work use this information to predict firm exits, but they do not seperate structural from crisis-induced employment loss or examine how employment risk was distributed among firms that received wage support.

2.4 How employment loss unfolds during a crisis

Employment loss occurs through two unique channels. Firms may reduce employment while continuing to operate, or they may exit the market altogether. These channels have different implications for workers, productivity and the allocation of public support. Davis, Haltiwanger and Schuh (1996) show that labour markets are characterised by substantial gross job destruction alongside job creation, highlighting the importance of analysing employment dynamics at the firm level. Because wage subsidy programmes aim to preserve both firms and jobs, evaluating their effects requires accounting for employment losses arising from both workforce reductions and firm exit.

2.5 The gap this study fills

Previous research provides each of these pieces separately. Wage subsidies address a genuine market failure, and broad support helped stabilize the economy during the acute phase of the crisis. Structural weakness and crisis shocks are recognised as distinct sources of employment risk. Firm characteristics such as age and size carry predictive signals for firm exits, and employment loss occurs through both firm exit and employment shrinkage.

The debate instead focuses on the consequences of broad support, including its cost and its effects on productivity, rather than on whether employment risk could have been identified while support was being allocated.

What the literature has not done is bring these elements together for the firms that actually received support. It has not distinguished, using only information available during programme implementation, how much expected employment loss reflects underlying structural vulnerability and how much is attributable to the crisis, nor related it to the allocation of emergency support.

Building on the literature, I address that gap by developing a framework that distinguishes expected structural from crisis-induced employment loss using administrative information available during programme implementation. Applying the framework to recipients of the Dutch NOW programme makes it possible to compare where support was allocated with where predicted crisis-related employment risk was concentrated, using only the information available when funding decisions were made.

3. Methodology

3.1 Overview

The methodology proceeds in four steps. The first estimates firm-level employment risk from historical LISA data by training two models: one predicting firm exit over a three-year horizon and one predicting employment shrinkage over a one-year horizon. The second applies these models to Round 1 NOW recipients to decompose expected job loss into two components: a structural baseline, derived from firm characteristics to represent underlying risk absent a crisis shock, and a crisis-induced deviation, derived from reported revenue losses and a crisis dummy to capture the additional risk triggered by the pandemic. The third translates these expected job losses into expected government costs using sector-average wages, statutory unemployment (WW) replacement rates, and UWV unemployment-duration records, comparing this expected treasury exposure against the Round 1 NOW subsidies actually disbursed. The fourth applies the exit model to all firms receiving NOW across rounds 1 to 8 to evaluate which of three rankings best corresponds to observed firm exits by 2022: one based on structural characteristics, one on reported revenue loss, and one combining the two.

3.2 Model Specification

The three questions all rest on two underlying prediction models, specified here. Employment risk is estimated from the LISA employment register, covering 2000 to 2016, using two logistic regression models that share the same specification but differ in their dependent variable. For firm i in sector s at time t:

$$Logit(P(Y_{\mathrm{ist}} = 1)) = \alpha + \beta_1 X_{\mathrm{i}} + \beta_2 TWV_{\mathrm{st}} + \beta_3 D_{\mathrm{t}} + \beta_4 (D_{\mathrm{t}} \times X_{\mathrm{i}}) + \gamma_{\mathrm{s}} + \delta_{\mathrm{p}} + \varepsilon_{\mathrm{ist}}$$

where X_i represents firm characteristics, TWV_{st} is the time-varying value-added shock (replaced by revenue loss at deployment), D_t is the crisis dummy, and γ_s and δ_p are sector and province fixed effects.

In the exit model, Y_{ist} equals one if the firm closes within three years. In the shrinkage model, Y_{ist} equals one if a surviving firm loses at least one employee within one year. The exit model omits the crisis dummy and its interactions because the crisis dummy already captures much of the effect reflected in reported revenue loss at deployment (see Appendix G).

Logistic regression is chosen to obtain an interpretable model that can be extrapolated beyond the historical training data, rather than to maximize predictive accuracy. Tree-based models cannot extrapolate to combinations outside those observed during training, and since the COVID-19 revenue losses observed during program implementation are substantially larger than the historical shocks used for estimation, a logit's continuous relationship can be evaluated for these larger losses where a tree cannot.

A hazard model could account for more exact timing of firm exits or employment adjustment. However, the objective of this study is to decompose expected employment loss over a fixed prediction horizon and compare aggregate outcomes across groups of firms rather than

predict the exact timing of individual events. A logistic regression is therefore sufficient enough.

Both models use the same pre-crisis structural predictors, all recorded in administrative records available for all firms: firm size (log employees), firm age (4 age groups), 1st-digit sector, province, and workforce composition (full-time share, female share and female full-time share). The estimation sample is restricted to firms with at least two employees. Further sample choices are described in Appendices D and F.

Revenue loss is not observed in the training data, since firms only reported it once NOW existed. The historical models therefore train on a sector-level proxy: the yearly change in each sector's value added (TWV), measured at parent-sector level and demeaned by the sector's long-run growth so it reflects below-trend shocks. This proxy is not collinear with the sector fixed effects. The static sector variables capture long-run average differences between industries, while TWV captures time-varying negative deviations from a sector's own average. Appendix A discusses the proxy and its relationship with reported revenue loss. The years 2009, 2010, 2012, and 2013 are marked as crisis years, with the crisis dummy interacted with sector, firm size, firm age, and full-time share. A single TWV coefficient is estimated without firm-characteristic interactions, so revenue loss also enters without interaction, avoiding overfitting the few crisis observations (Appendix B).

During programme deployment, the crisis marker is set to one for every NOW recipient and the sector-level TWV proxy is replaced by each firm's reported quarterly revenue loss. The reported loss is multiplied by the coefficient estimated for TWV, making revenue loss the firm-specific crisis measure used by the framework. Since revenue loss is not directly observed in the data, it is reconstructed from NOW advance payments (Appendix C).

3.3 Expected Job Loss

This section describes how the predicted probabilities from the exit and shrinkage models are converted into the employment-loss measures used throughout the remainder of the analysis.

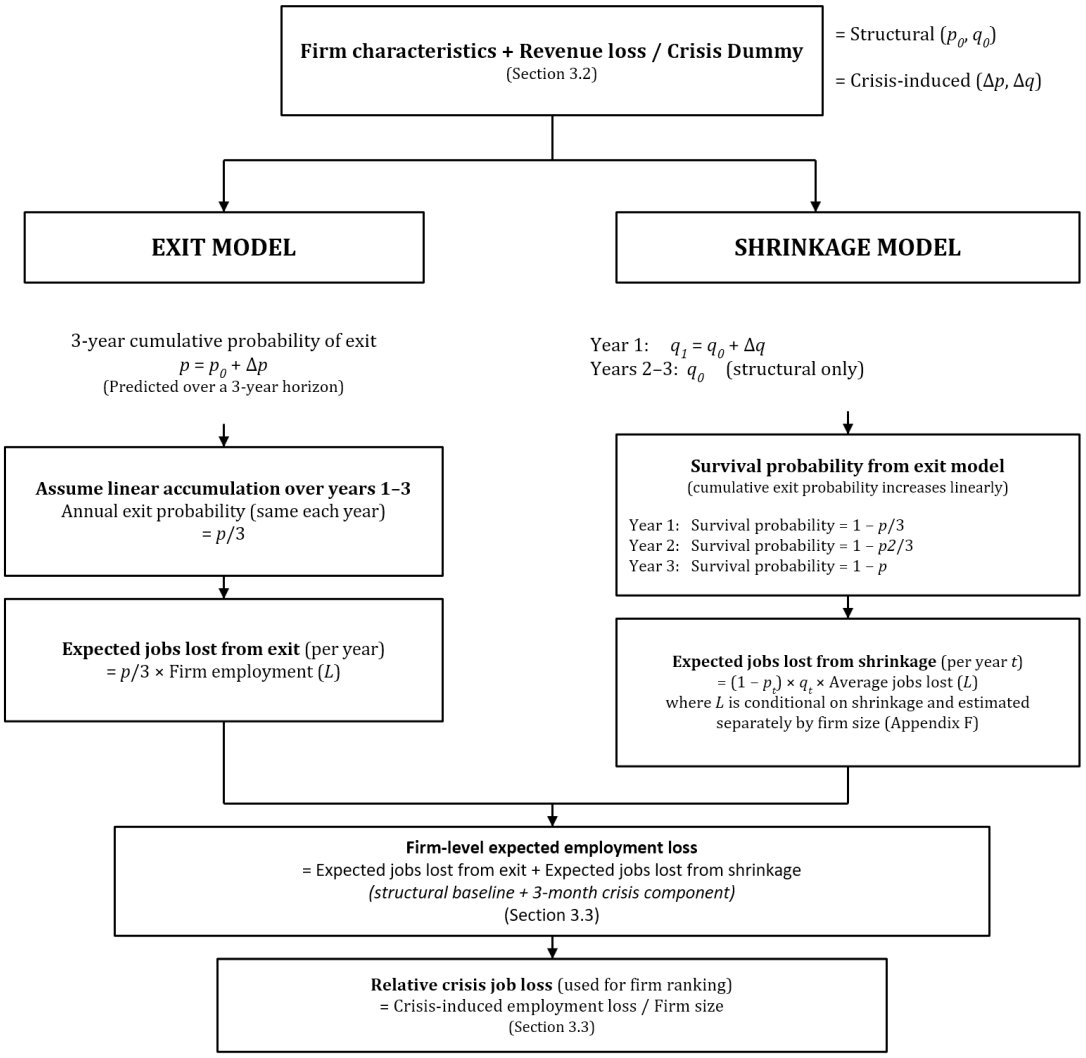


Figure 3.1. *The exit and shrinkage models, each decomposed into structural and crisis-induced components, combine to estimate firm-level expected employment loss. In Sections 5.1–5.3, the crisis-induced component is scaled to the approximately three-month duration of Round 1 NOW (see Appendix E).*

Figure 3.1 summarises the framework. Firm characteristics and observed revenue loss enter both models. The exit model predicts a firm's cumulative probability of exit over a three-year horizon, while the shrinkage model predicts the probability that a surviving firm reduces employment. In both models, the structural component represents predicted employment loss in the absence of a crisis shock, while the crisis-induced component captures the additional employment loss associated with the observed revenue shock.

For the exit model, the predicted probability is

$p = p_0 + \Delta p$

where p_0 is the structural probability of exit and Δp the additional probability attributable to the crisis. Expected employment loss from exit equals the predicted exit probability multiplied by firm employment. Because the model output is a probability, these are probabilistic quantities, not forecasts for individual firms: a single firm's expected loss may be fractional, such as 0.2 of a job, and is interpretable only in aggregate, where individual expectations sum to meaningful totals. All expected employment losses in this paper, structural, crisis-induced, and the residual employment not expected to be lost, are interpreted at this aggregate level.

Because the model predicts cumulative exit over three years, the cumulative exit probability is assumed to increase linearly across the three years (see Appendix D). This provides an annual survival probability for each year, used when calculating expected losses from within-firm shrinkage.

For the shrinkage model, the crisis shock is assumed to affect only the first year, so the first-year shrinkage probability is

$$q_1 = q_0 + \Delta q,$$

while the second and third years use the structural probability q_0 only. Expected employment loss from shrinkage is

$$(1 - p_t) \times q \times L,$$

where $1 - p_t$ is the probability the firm remains in operation in year t, q is the annual probability of shrinkage, and L is the average number of jobs lost conditional on shrinkage, estimated separately by firm size (Appendix F). This weighting prevents double-counting: a firm already expected to have exited cannot also shrink, so weighting shrinkage by the probability the firm still operates keeps the two channels from together exceeding a firm's employment.

The underlying models are estimated on annual outcomes, whereas Sections 5.1 to 5.3 evaluate the roughly three-month duration of Round 1 NOW. The estimated crisis component is therefore divided by four. Reported crisis-induced losses should in turn be read as the expected effects of a three-month crisis, while the structural component represents the longer-run loss expected absent the COVID-19 shock. Because revenue losses worsen in later rounds as firms run down reserves, Round 1 understates full crisis severity; the rationale for using it is discussed in Appendix E.

To ensure firms are ranked by crisis intensity rather than size, combined crisis-induced employment loss is normalised by total firm employment. Firms are then sorted by this relative crisis job loss measure and allocated to five equally sized tiers, from Tier 1 (lowest crisis job loss rate) to Tier 5 (highest).

3.4 Expected Unemployment Benefit Costs

This section describes how predicted employment loss is translated into expected unemployment-benefit (WW) costs under a no-intervention scenario and compared with Round 1 NOW expenditure.

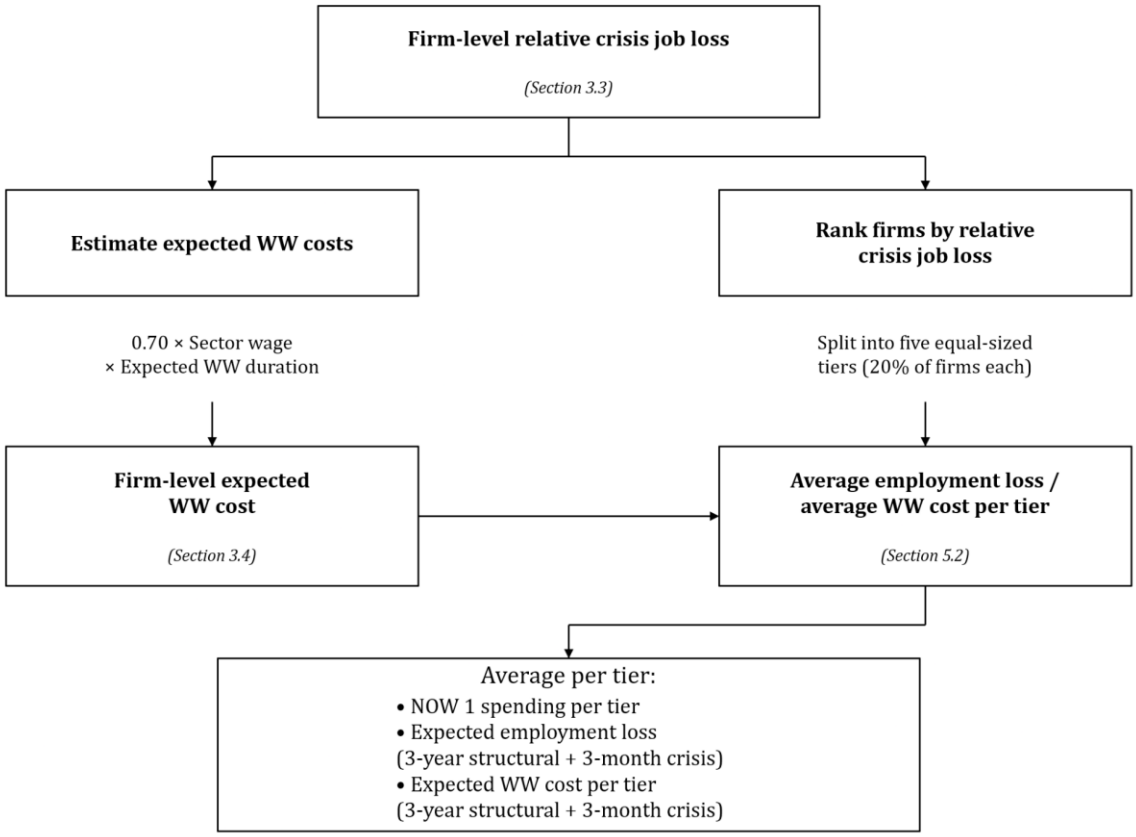


Figure 3.2. *From firm-level expected employment loss, the left branch converts each firm's loss into an expected WW cost while the right branch ranks firms into five tiers, which are combined to give average NOW spending, expected employment loss, and WW cost per tier.*

The second question concerns how predicted employment loss can be expressed as public cost. Figure 3.2 shows how each firm's expected employment loss is translated into the unemployment cost it would impose on the government under a no-intervention scenario, averaged within risk tiers and set against the NOW spending those firms received. The analysis focuses on expected WW costs as one important public cost consequence of employment loss. It does not attempt to estimate the program's full social or public costs and benefits.

For every job expected to be lost, the government carries two monthly costs for as long as the worker stays unemployed: the unemployment benefit it pays (70 percent of prior wage) and the labour-tax revenue it no longer collects (around 37 percent, the Netherlands' average tax wedge on labour; OECD, 2025). Expected cost is these per-month costs multiplied by the expected jobs lost and by sector-specific unemployment duration, split into a structural and a crisis component:

$$Expected\ Cost\ =\ Expected\ Jobs\ Lost\ \times\ (W_comp\ \times\ \varphi\ \times\ 0.70\ +\ W_comp\ \times\ 0.37)\ \times\ Du$$

where W_comp is the sector's average monthly wage, Du its average unemployment duration, and φ caps wages above the benefit ceiling. Durations are estimated from UWV administrative data; the full derivation, the single-rate tax-wedge assumption, and the wage-base treatment are in Appendix K.

3.5 Do Predicted Risk Rankings Correspond to Subsequent Firm Outcomes?

This section describes how the framework evaluates whether administrative information available during programme implementation distinguished firms that later exited from those that survived.

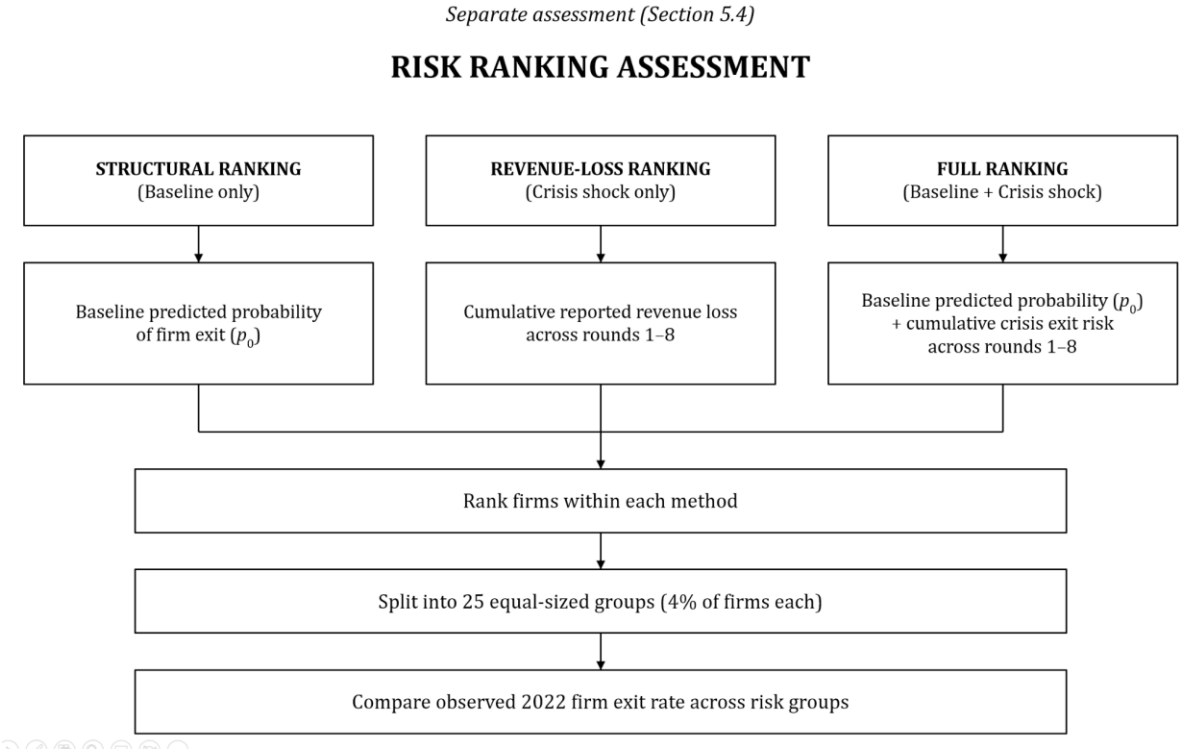


Figure 3.3 Ranking procedure for the exit model: firms are ranked by predicted exit probability, split into 25 equal-sized groups (4% each), and their predicted exit compared against exit observed by 2022.

The third question examines whether firms that later exited can be distinguished in advance from those that survived, conditional on receiving NOW support, and which risk ranking does this most effectively. If firms assigned to progressively higher predicted-risk groups subsequently exhibit higher exit rates, then the information needed to isolate firms that remained vulnerable despite support was already available before outcomes were observed. The analysis evaluates the information content of alternative rankings rather than the policy responses they might justify.

The test uses firms that received NOW across rounds 1 to 8 and the exit model alone. Each firm's predicted exit probability is used to rank all firms, which are split into 25 equal-sized groups and compared with the exit actually observed by 2022. Twenty-five groups, rather than the five tiers used in the cost analysis, provide a more detailed view of how observed

exit rates change across the predicted risk ranking. This focuses on the relative ordering of firms instead of probability thresholds and allows consistent comparisons across sectors.

Three rankings are compared. The structural ranking uses each firm's baseline exit probability from pre-crisis characteristics alone. The revenue-loss ranking uses cumulative reported revenue loss across all NOW rounds. The full ranking starts from the structural exit probability and adds the crisis-induced increase in exit risk for each round, giving a cumulative predicted exit risk over the programme. Comparing the three isolates the contribution of each source: if the structural ranking fails while the revenue-loss ranking succeeds, the predictive signal comes from the crisis shock rather than pre-crisis fragility, and if the full ranking outperforms either input alone, combining structural and crisis-specific information adds predictive value.

4. Data

4.1 LISA Employment Register

The primary data source is the LISA employment register (LISA, n.d.), a Dutch administrative dataset covering virtually all firms operating in the Netherlands. LISA records firm-level employment, sector classification, province, size category, and year of founding for each firm annually. The training panel covers 2000 to 2016, yielding approximately 10 million firm-year observations after restricting to firms with at least two employees. Sectors A (Agriculture, Forestry and Fishing) and B (Mining and Quarrying) are not included in the GOLD training extract and consequently do not appear in the estimation sample. However, these sectors account for only a small share of Round 1 NOW recipients and expenditure (Smits et al., 2022), suggesting that their exclusion is unlikely to materially affect the results. The 2019 LISA wave provides pre-crisis firm characteristics for the deployment sample. Both are accessed through the FIRMBACKBONE infrastructure (FIRMBACKBONE, n.d.), which links LISA observations across years using a consistent firm identifier.

4.2 NOW Administrative Data

NOW subsidy data are originally provided by UWV (UWV, n.d.) and cover all eight rounds of the programme. The data record the advance payment received by each firm in each subsidy round. Through the FIRMBACKBONE infrastructure (FIRMBACKBONE, n.d.), these records are assigned a consistent firm identifier that enables linkage to the LISA employment register, using the 2019 LISA wave where available and subsequently matching remaining firms to the 2020, 2021 and 2022 LISA waves.

4.3 Data Linkage and Sample Construction

NOW recipients are linked to the LISA employment register via the FIRMBACKBONE identifier, first to the 2019 wave and then, where unmatched, to the 2020, 2021, and 2022 waves. Of 81,339 Round 1 recipients, 51,044 (62.8 percent, and 69.2 percent of Round 1 spending) are matched. Matched firms receive larger average subsidies than unmatched ones (€65,719 versus €51,477), indicating a mild size bias in linkage. After excluding firms with missing employment or implausible founding years, the final deployment sample

contains 194,161 firm-round observations across the eight rounds. Because unmatched firms are smaller and reconstruction error is largest among the smallest firms (Appendix C), the sampling bias and the measurement error run in the same direction: the firms that drop out are also the ones measured least reliably. The analytical sample is hence better measured than the full population, at the cost of underrepresenting the smallest firms.

Sector	N firms	Median jobs	Median age	Mean rev. loss <8 (%)	Mean rev. loss >8 (%)	Mean rev. loss (%)	Total NOW budget (€M)	% of total budget
G	13,809	6.0	3	49.3%	44.2%	48.1%	848.7	20.8%
C	4,625	11.0	2	38.0%	33.5%	37.2%	622.0	15.3%
N	3,228	5.5	2	65.7%	58.4%	64.5%	468.2	11.5%
J	2,116	6.0	1	43.6%	38.2%	42.7%	431.3	10.6%
I	8,226	8.0	2	70.4%	60.7%	68.8%	430.1	10.6%
M	5,288	6.0	2	44.4%	37.7%	43.1%	375.0	9.2%
H	1,985	9.0	1	42.5%	34.5%	41.2%	325.6	8.0%
R	2,083	7.0	2	61.0%	54.4%	60.1%	161.9	4.0%
F	2,130	9.0	2	37.1%	29.9%	35.6%	118.4	2.9%
Q	2,820	6.0	2	42.6%	34.3%	41.4%	102.6	2.5%
P	983	5.0	2	45.4%	40.9%	44.5%	49.3	1.2%
S	1,662	4.0	3	58.8%	48.3%	56.5%	46.4	1.4%
K	372	6.0	2	33.5%	26.6%	32.2%	27.2	0.7%
L	635	5.0	3	37.5%	33.4%	36.9%	19.9	0.5%
E	110	9.0	2	34.8%	37.3%	35.5%	6.5	0.2%
D	13	5.0	2	41.7%	15.3%	37.6%	0.9	0.0%
O	6	108.0	2	8.7%	0.0%	7.3%	0.4	0.0%
Total / Avg	**50,091**	**12.7**	**2.1**	**44.4%**	**36.9%**	**43.1%**	**4034.4**	**100%**

Table 4.1. Round 1 NOW applicants by sector. *N* is the number of firms in the matched sample. Mean revenue loss is reconstructed from NOW advance payments. Total NOW budget reports matched Round 1 expenditure (€ million), and % of total budget each sector's share of the matched total (€4.07 billion). Sectors are ordered by total NOW budget. The nine largest market sectors (C, F, G, H, I, M, N, R and S) form the estimation sample for the crisis component of the exit and employment-loss models. Excludes sectors A and B and firms with missing values.

Two features of Table 4.1 matter for what follows. First, expenditure is highly concentrated: wholesale and retail (G), manufacturing (C), and administrative services (N) together account for roughly half of the €4.07 billion in matched Round 1 spending. Second, reported revenue losses vary widely across sectors, from around 36 percent in construction (F) to nearly 69 percent in hospitality (I), foreshadowing the crisis-exposure differences that drive the later results. The crisis component of both models is estimated on the nine largest market sectors only(Appendix B).

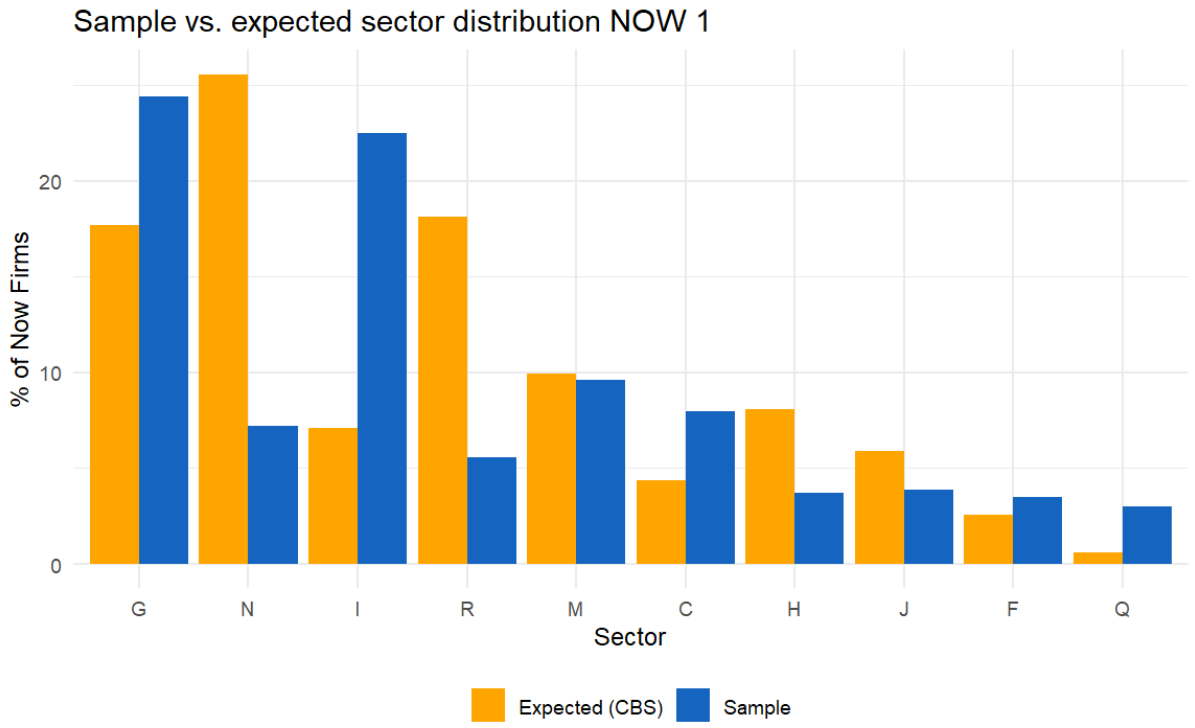


***Figure 4.2**. Matched round 1 sample versus expected NOW round 1 sector distribution (CBS estimates); the sample broadly tracks the population, with gaps concentrated in small-firm sectors.*

Figure 4.2 compares the sector distribution of the matched sample against the expected distribution of NOW round 1 recipients implied by the CBS participation estimates in Smits et al. (2022). These gaps likely reflect differences in how consistently firms can be linked to the LISA register, and the resulting sector imbalances should be kept in mind when interpreting sector-level results.

4.4 Supplementary Data

Sector-level value added growth (TWV) is obtained from CBS StatLine (CBS, n.d.-a) at the parent sector letter level for 2000 to 2020, and is used as the crisis shock measure in model training. Sector average wages including employer social contributions are obtained from CBS National Accounts (CBS, n.d.-b) at the two-digit SBI level. Expected WW unemployment benefit durations are obtained from UWV administrative records by sector (UWV, n.d.). A full wage and WW data overview can be seen in Appendix J. Firm exit by 2022 is determined from the 2022 wave of FIRMBACKBONE (FIRMBACKBONE, 2022),

which incorporates the LISA employment register (LISA, 2016) used for the 2000–2016 training data.

5.Results

5.1 Risk tiers and Jobs lost

Both models assign each firm a predicted probability that splits into structural and crisis-induced components. This section first reports the decomposition for firm exit (5.1.1) and employment shrinkage (5.1.2), before combining both channels into expected employment loss and relative crisis job loss tiers(5.1.3).

5.1.1 Structural and Crisis Components of Exit Risk.

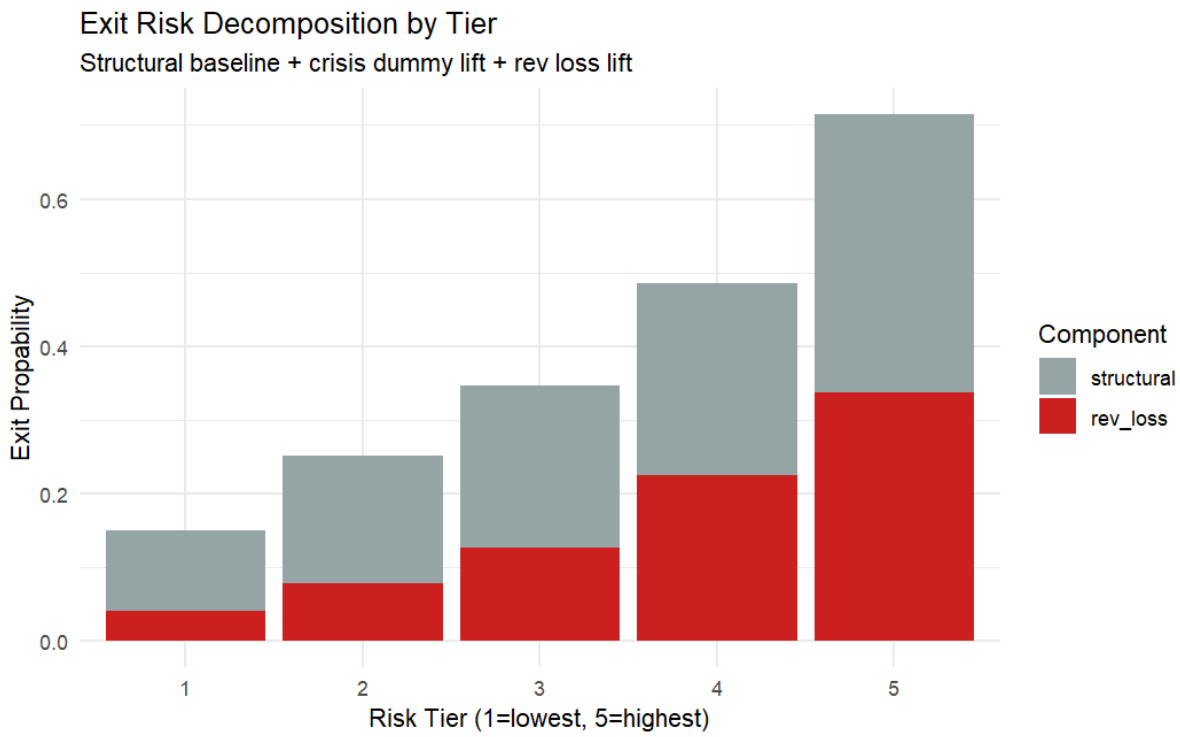


Figure 5.1. *Average predicted exit probability by predicted exit-risk tier. The x-axis groups Round 1 NOW recipients into five equal-sized tiers ranked by total predicted exit probability (1 = lowest, 5 = highest). The y-axis reports the average predicted exit probability within each tier. Each bar is decomposed into the structural component predicted from pre-crisis firm characteristics (grey) and the additional crisis-induced component associated with reported revenue loss (red).*

The composition shifts across the ranking: structural risk dominates the lower tiers, while the revenue-loss component approaches half of predicted risk in Tier 5. Higher-risk firms therefore combine greater underlying vulnerability with larger crisis shocks. The structural component assigns higher exit risk to smaller, younger firms with more part-time and fewer female employees, and achieves moderate out-of-sample discrimination (AUC = .69). The crisis dummy is omitted because it is collinear with revenue loss (Appendix G).

5.1.2 Shrink Risk Tier

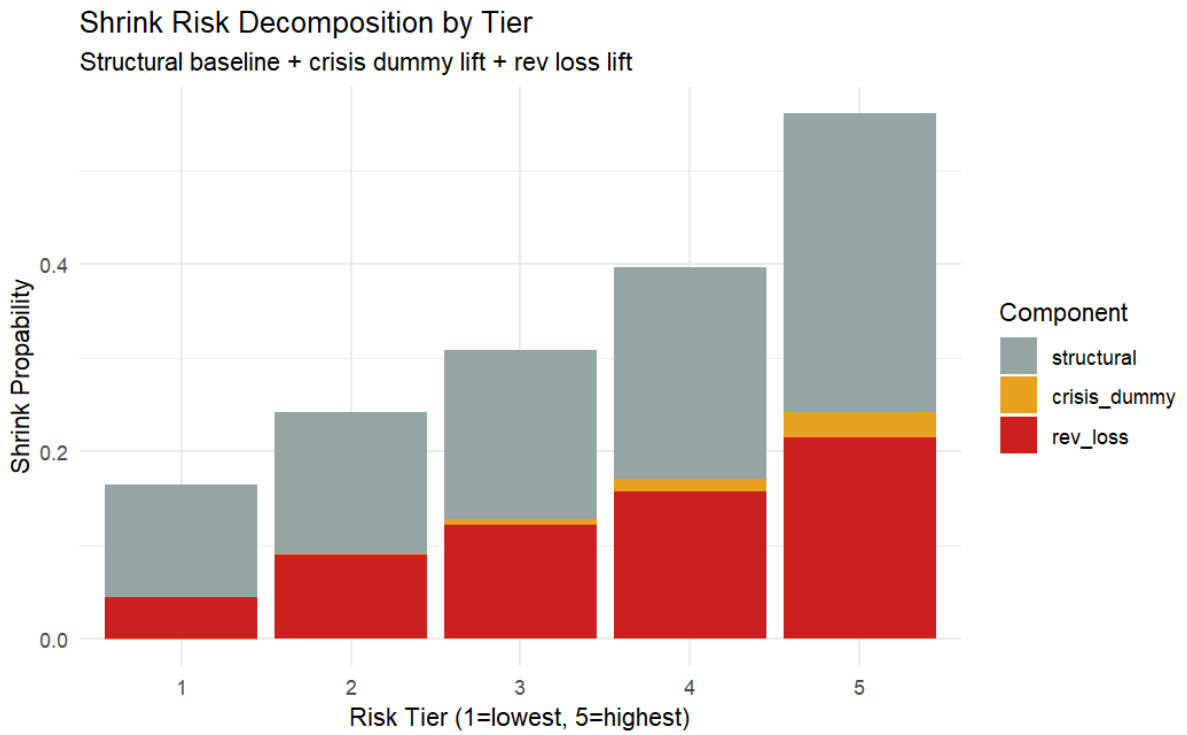


Figure 5.2. *Average predicted employment-shrinkage probability by predicted shrinkage-risk tier. The x-axis groups Round 1 NOW recipients into five equal-sized tiers ranked by total predicted shrinkage probability (1 = lowest, 5 = highest). The y-axis reports the average predicted shrinkage probability within each tier, decomposed into structural, crisis-dummy and revenue-loss components.*

The shrinkage model assigns each firm a predicted probability of reducing employment, split into a structural component and two crisis components: a common crisis-dummy effect and an additional revenue-loss effect. Firms are ranked by total predicted shrinkage probability and grouped into five equal-sized tiers (Figure 5.2).

Predicted shrinkage risk rises from about 0.15 in Tier 1 to 0.50 in Tier 5. The composition also changes across the ranking. In Tier 5, for example, the average predicted probability of employment shrinkage is 56%: around 30 percentage points reflect structural risk, while the remaining 26 percentage points are crisis-induced, comprising about 23 points from reported revenue loss and 3 from the common crisis effect. Structural risk dominates the lower tiers, whereas the revenue-loss component becomes the main source of crisis-induced risk in the highest tiers. The common crisis effect remains small throughout.

Unlike the exit model, the shrinkage model retains both the crisis dummy and reported revenue loss. The structural component predicts greater employment shrinkage for larger firms and those with a lower full-time share. The crisis dummy captures an economy-wide increase in shrinkage risk independent of reported revenue loss, while its interactions shift that increase towards older, larger and more full-time-intensive firms. The structural model achieves moderate out-of-sample discrimination (AUC = .65). Coefficients and size-specific job-loss multipliers are reported in Appendix H and F.

5.1.3 Expected Job Loss in Absolute Terms

This section combines the predicted exit and employment-shrinkage probabilities into expected employment loss and introduces the measure of relative crisis job loss used throughout the remainder of the analysis.

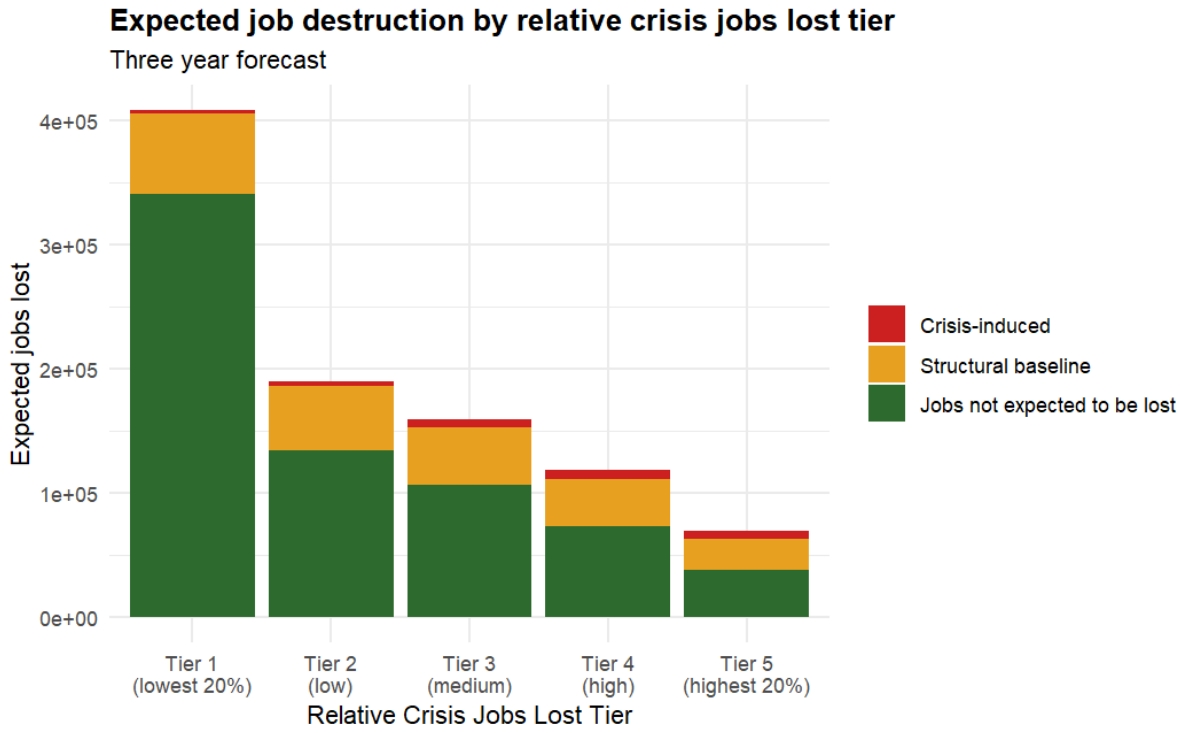


Figure 5.3. *Model-predicted employment outcomes over a three-year forecast horizon across firms grouped into five tiers by relative crisis-induced employment loss (x-axis). The y-axis reports the aggregated expected number of jobs in each tier, decomposed into retained employment (green), structural employment loss (orange), and crisis-induced employment loss (red).*

Figure 5.3 translates the predicted probabilities into expected employment loss over the three-year horizon. Total expected losses and the structural component concentrate in the lower tiers, which hold much larger firms. Crisis-induced losses, however, rise mostly across the tiers in absolute terms, even as total employment falls. Ranking firms by crisis loss relative to workforce hence also orders them close to absolute crisis exposure, isolating the firms most affected by the crisis rather than simply the largest.

5.2 Where did the support end up?

This section examines how Round 1 NOW expenditure was spread across those tiers and what employment each euro of expenditure was associated with, and explains why similar budgets were allocated despite large differences in the number of jobs.

Table 5.4 Jobs reached per €1 million of NOW spending, by relative crisis jobs tier

Tier	*Now Spending (€B)*	*Crisis / €1M*	*Structural / €1M*	*Safe / €1M*	*Total Jobs / €1M*
Tier 1 (lowest 20%)	*0.77*	*3*	*83*	*442*	*528*
Tier 2 (low)	*0.61*	*6*	*85*	*221*	*312*
Tier 3 (medium)	*0.98*	*6*	*48*	*109*	*163*
Tier 4 (high)	*0.78*	*10*	*48*	*94*	*152*
Tier 5 (highest 20%)	*0.90*	*8*	*28*	*42*	*78*

Table 5.4. *Expected employment composition associated with €1 million of Round 1 NOW expenditure by relative crisis job-loss tier. Values represent the aggregated firm-level expected numbers of crisis-induced, structural and safe jobs associated with each €1 million of observed NOW spending within each tier.*

The previous section showed that higher relative crisis job-loss tiers contain substantially fewer jobs overall. Despite this, Round 1 NOW expenditure remains relatively evenly distributed across the five tiers, ranging from €0.61 billion to €0.98 billion (Table 5.4). This reflects offsetting effects. Moving from Tier 1 to Tier 5, average reported revenue loss rises sharply (12 to 98 percent), increasing subsidy payments, while average firm size falls (around 41 to 7 employees) and average monthly wages decline modestly (€4,000 to €2,600), reducing the wage bill on which payments are based (Table 5.6). These offset one another, resulting in similar aggregate expenditure across the relative crisis job-loss ranking.

Table 5.4 expresses expenditure as the employment associated with each €1 million of observed NOW spending. Within each tier, the total expected jobs of each type are divided by total NOW spending in that tier. For example, suppose Firm A receives €2 million with 100 safe, 40 structural and 60 crisis-induced jobs, and Firm B receives €1 million with 20 safe, 10 structural and 20 crisis-induced jobs. Together, €3 million is associated with 120 safe, 50 structural and 80 crisis-induced jobs, that is, 40 safe, 16.7 structural and 26.7 crisis-induced jobs per €1 million. Applying this within each tier gives the values in Table 5.4.

The results show that each €1 million of NOW expenditure reaches progressively fewer jobs overall, reflecting the smaller firms in higher relative crisis tiers. However, the composition of those jobs changes substantially. Employment not expected to be lost declines sharply, while crisis-induced and structural employment account for an increasing share of the employment reached by each €1 million. Thus, although the absolute number of crisis jobs

per €1 million does not increase monotonically, expenditure becomes increasingly concentrated on employment the model predicts to be at risk, particularly by reducing the share of spending associated with jobs not expected to be lost.

5.3 Employment Loss and Expected WW Costs

This section expresses predicted employment loss as expected unemployment-benefit (WW) costs under a no-intervention scenario and compares these estimates with Round 1 NOW expenditure.

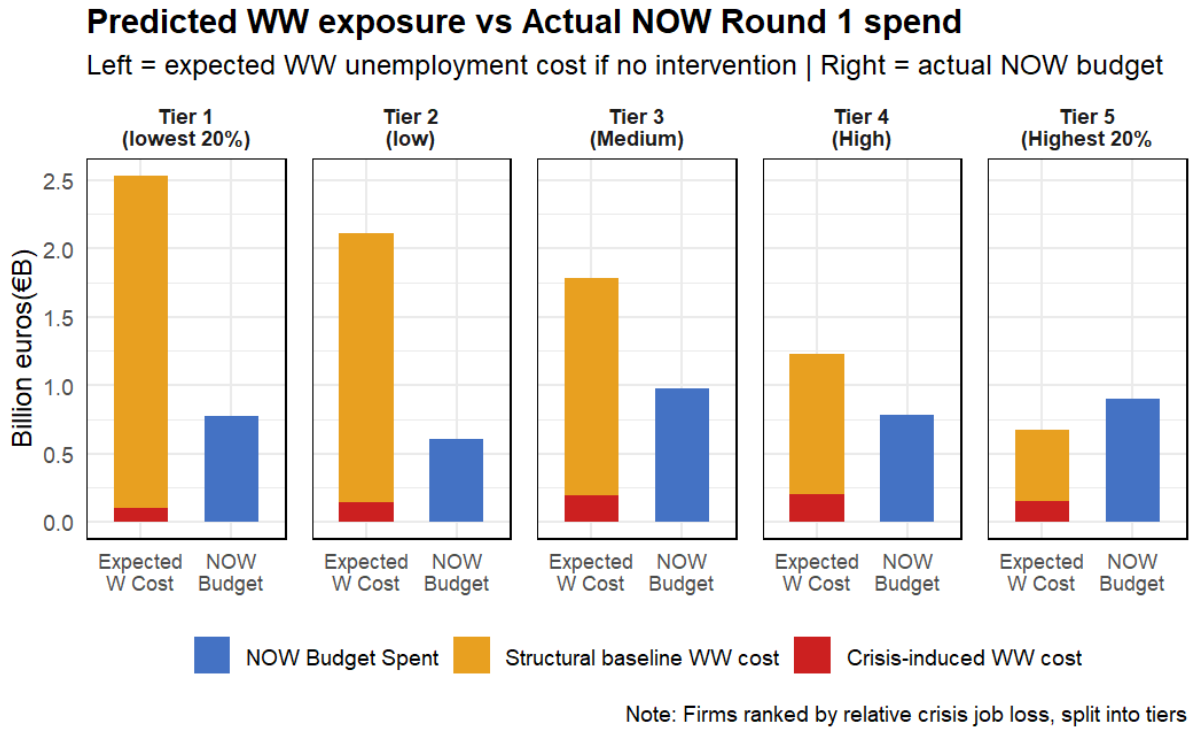


Figure 5.5. *Expected unemployment-benefit (WW) costs under a no-intervention scenario (left bars), decomposed into structural and crisis-induced components, compared with actual Round 1 NOW expenditure (right bars) across relative crisis job-loss tiers.*

Figure 5.5 compares predicted WW costs under the no-intervention scenario with actual Round 1 NOW expenditure. While NOW expenditure remains relatively evenly allocated across the relative crisis ranking, expected crisis-induced WW costs increase from Tier 1 to Tier 4 before declining in Tier 5. WW represents only one public-cost channel among several, so the comparison illustrates how predicted employment loss relates to one component of public costs rather than providing a complete assessment of programme costs and benefits. Table 5.6 explains the drivers of this pattern.

What Drives Expected WW Cost? By Combined Crisis Jobs lost Tier

Green = factor higher relative WW cost | Red = factor lower relative WW cost

Tier	*Dominant Sector*	*Avg Revenue Loss (%)*	*Avg Monthly Wage (€)*	*Avg Unemp. Duration*	*Avg Firm Size*	*Crisis-induced WW cost (€B)*	*Total WW Cost (€B)*
Tier 1 (Lowest 20%)	*G (Retail)*	*12.3%*	*€4,026*	*8.83*	*40.8*	*€0.099B*	*€2.578B*
Tier 2 (Low)	*G (Retail)*	*25.3%*	*€3,978*	*8.89*	*19.0*	*€0.144B*	*€2.165B*
Tier 3 (Medium)	*G (Retail)*	*41.6*	*€3,658*	*8.79*	*15.9*	*€0.192B*	*€1.841B*
Tier 4 (High)	*I (Hospitality)*	*72.4*	*€3,251*	*8.62*	*11.9*	*€0.202B*	*€1.263B*
Tier 5 (Highest 20%)	*I (Hospitality)*	*98.4%*	*€2,579*	*8.26*	*7.0*	*€0.157B*	*€0.719B*
All tiers	***-***	***50.0%***	***€3,498***	***8.68***	***18.9***	***€0.794B***	***€8.566B***

Table 5.6. *Drivers of expected WW costs by relative crisis job-loss tier. Crisis-induced WW cost reports the additional expected unemployment-benefit cost attributable to COVID-19, while Total WW cost includes both structural and crisis-induced components.*

Expected WW costs depend on predicted job losses together with sectoral wages and expected unemployment duration, whereas NOW expenditure depends primarily on reported revenue loss and firms' wage bills. Moving up the ranking, firms become smaller and wages decline as the dominant sector shifts from retail to hospitality. Lower wages reduce the expected WW cost associated with each lost job, causing expected crisis-induced WW costs to peak in Tier 4 rather than Tier 5 despite the higher relative concentration of crisis-induced employment loss.

This result complements the findings from Section 5.2. Although each euro of NOW expenditure became increasingly associated with crisis-induced employment loss and less associated with employment predicted to survive, this does not translate proportionally into expected WW costs. From the perspective of WW expenditure alone, the potential reduction in expected benefit costs is consequently not necessarily greatest in the highest relative crisis tier. Instead, the lower tiers represent substantially greater expected WW exposure because they contain larger and higher-paid firms. Whether this translates into lower overall treasury costs remains ambiguous, since this requires assumptions about intervention effectiveness, and broader fiscal and economic consequences of programme targeting which are outside the scope of this study.

5.4 Do Predicted Risk Rankings Correspond to Subsequent Firm Exits?

This section examines whether the information used by the framework also distinguishes firms that subsequently exited from those that survived.

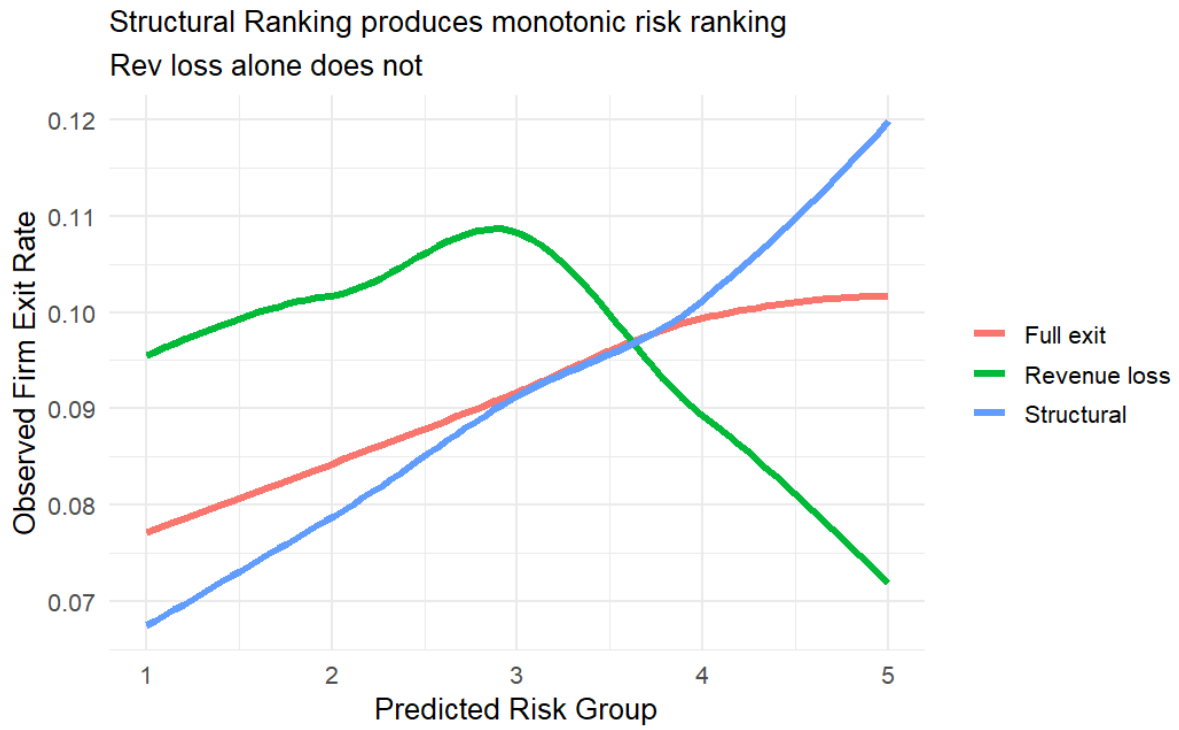


Figure 5.7. *Observed firm exit rates by 2022 across 25 predicted-risk groups under three alternative rankings: structural risk, reported revenue loss and the full exit-risk model. The x-axis groups firms into 25 equal-sized predicted-risk for each ranking method, groups from lowest to highest predicted exit risk. The y-axis reports the observed firm exit rate by 2022.*

Figure 5.7 compares how well the three approaches distinguish firms that later exited from those that survived. A ranking is informative if progressively higher predicted-risk groups also exhibit progressively higher observed exit rates.

Structural risk provides the clearest ordering of subsequent firm exits. Observed exit rates increase almost monotonically across the structural groups while exhibiting the greatest separation between the lowest- and highest-risk firms. By contrast, reported revenue loss alone produces a non-monotonic pattern, with firms in the highest revenue-loss groups exhibiting lower exit rates than firms with more moderate reported losses. Adding revenue loss to the structural ranking reduces rather than improves discrimination, producing a flatter relationship between predicted risk and observed firm exits.

This aggregate pattern masks sectoral differences. Repeating the comparison within the largest NOW sectors shows that the structural ranking remains the most consistent predictor across industries, whereas the contribution of reported revenue loss varies substantially. In some sectors it adds little, while in hospitality (sector I) it ranks firms in the opposite direction to structural risk, suggesting that the two capture different dimensions of firm vulnerability. The sector-level results are reported in full in Appendix I.

6. Discussion

6.1 Main Findings and Contribution

Using only information available before outcomes occurred, the framework consistently isolated predicted structural from crisis-induced employment loss across both the exit and shrinkage channels. Administrative data available during the pandemic could therefore separate the two sources of employment risk.

Round 1 NOW expenditure was distributed relatively evenly across the relative crisis job-loss ranking, but the composition of employment reached by each euro differed systematically. Structural employment loss remained substantial throughout, while each euro became associated with a higher concentration of crisis-induced loss and a smaller share of employment predicted to survive. Expressed as expected WW costs, however, a different pattern emerges. Although the highest relative crisis tiers contain the greatest concentration of crisis-induced employment loss, the lower tiers are associated with substantially higher total expected WW costs under a no-intervention scenario because they contain larger and higher-paid firms. This suggests that the firms where the crisis is most concentrated are not necessarily those associated with the greatest potential reduction in WW expenditure, while the implications for overall public costs remain outside the scope of this study.

Structural firm characteristics produced the clearest ordering of subsequent firm exits, while reported revenue loss alone ranked firms less consistently and, when combined with structural characteristics, did not improve discrimination. Sector-level comparisons were largely inconclusive, with all three rankings performing unevenly across industries.

Together, these findings suggest that administrative information available during programme implementation could be used to seperate structural from crisis-induced employment risk, relate this to the observed allocation of NOW expenditure, and estimate one important channel of expected public costs under a no-intervention scenario. This does not imply NOW was poorly designed: the programme was implemented under exceptional time pressure, and broad eligibility was likely the most practical response. Whether such information could realistically have been incorporated under those constraints is a separate policy question.

6.2 Limitations

Estimating Crisis-Induced Employment Loss

The principal limitation concerns the estimation of crisis-induced employment loss. The crisis response is estimated from historical downturns, primarily the 2008 financial crisis and subsequent recessions, and applied to COVID-19. Because the framework captures firms' average employment adjustment across past adverse economic conditions rather than distinguishing between different types of shocks, the estimated crisis effects may not transfer perfectly to the pandemic. This is reflected in the different roles of the crisis components across the two models: for firm exit the common crisis effect is largely captured by the crisis dummy, whereas for employment shrinkage the crisis dummy and reported revenue loss contain complementary information.

Employment Adjustment

The framework captures employment loss through firm exit and reductions in firm employment, but not other forms of labour adjustment within surviving firms, such as reduced working hours or changes in contract composition. This means employment-loss estimates, only captures part of firms' labour-market adjustment.

Aggregate Interpretation

The framework is intended for aggregate analysis rather than precise firm-level prediction. Firms with similar reported revenue losses may respond differently because they differ in unobserved financial characteristics such as liquidity or debt-servicing capacity. At the individual-firm level these differences introduce prediction error. When firms are aggregated, however, much of this variation averages out, making the framework more informative for comparing groups of firms than for predicting outcomes for any single firm (Banerjee & Kharroubi, 2020).

6.3 Future Research

Future work could apply the framework to alternative crises in which untreated firm outcomes are observable, distinguish between different types of economic shocks, and estimate sector-specific crisis responses. Incorporating richer firm-level financial information, such as liquidity or leverage, could allow the estimated crisis response to vary across firms rather than assuming a common relationship with reported revenue loss. Finally, expected treasury costs could be refined using more detailed administrative information on taxes and benefit entitlements.

Appendix A - Turnover Change as a Proxy for Value Added Change

The exit and employment models are trained on sector-level value added change (TWV) as the crisis shock measure. Turnover measures a firm's total sales, whereas value added measures the value created after subtracting the cost of intermediate inputs:

$$Turnover = Sales$$

$$Value\ added = Sales - Intermediate\ inputs$$

Both measures therefore reflect changes in underlying economic activity, although value added excludes fluctuations driven purely by changes in purchased inputs. Because consistent historical value-added data are available across sectors and years, TWV is used to estimate the historical crisis models. At the time of intervention, however, value added is not yet observable. Policymakers instead observe firms' declared quarterly revenue losses through NOW applications.

This appendix examines whether sector-level turnover change provides a reasonable proxy for sector-level value added change, thereby assessing the validity of applying model coefficients estimated using TWV to firm-declared revenue losses.

To examine the relationship between the two measures, I regress 2020 sector-level value added change on sector-level turnover change across 51 two-digit SBI sectors using CBS StatLine data.

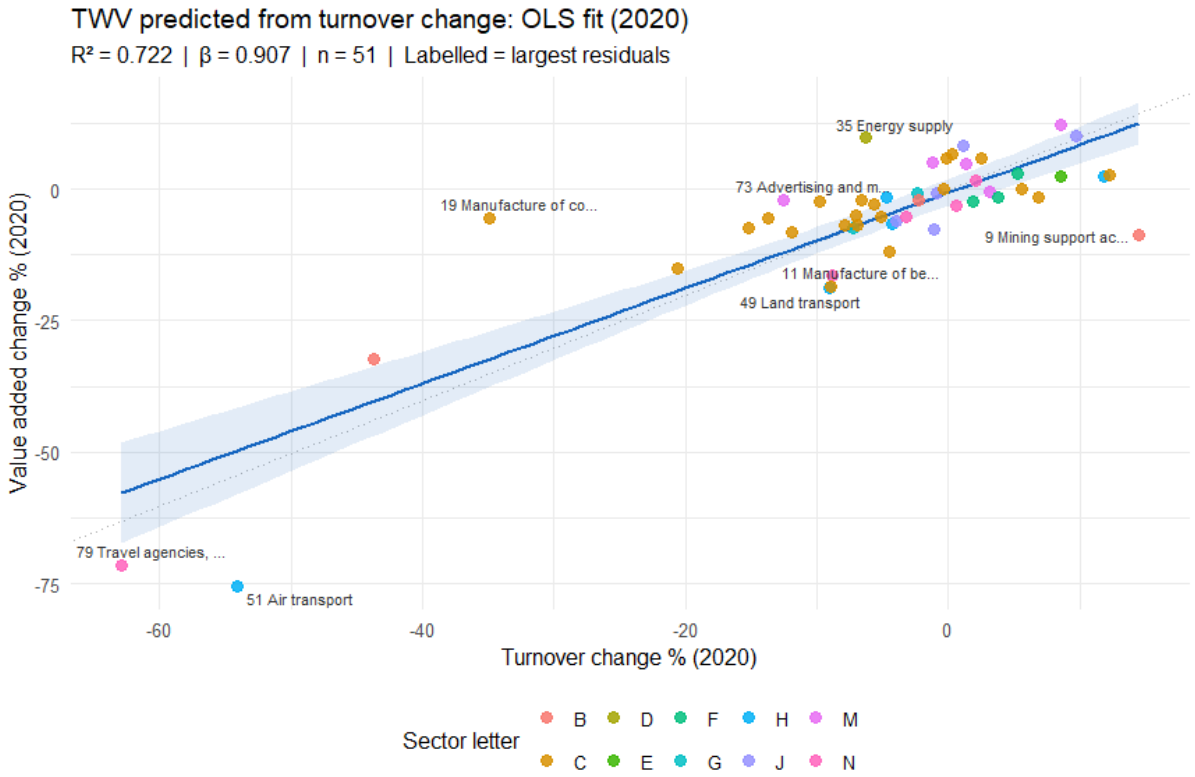


***Figure A1**. Sector-level value added change (TWV) against turnover change in 2020 across 51 two-digit SBI sectors, with the OLS fit ($R^2 = 0.722$, $\beta = 0.907$); the two move closely together, with the largest residuals labelled.*

Regression analysis

The pooled OLS regression yields $R^2 = 0.722$ with $\beta = 0.907$, indicating that a one percentage point decline in turnover is associated with approximately a 0.907 percentage point decline in value added. The two measures move proportionally for most sectors, with the intercept of -0.694 being negligible in practice.

Adding sector letter fixed effects raises R^2 to 0.808, suggesting that broad sector categories account for some remaining systematic differences in the turnover-TWV relationship - likely reflecting structural differences in input intensity and cost rigidity across sector groups.

Within-sector patterns

The relationship is strongest and most linear in the sectors most affected by COVID-19 and most heavily represented in NOW applications transport and storage (H), temporary employment agencies and travel (N), and wholesale and retail trade (G)(Smits 2022). Sectors showing weaker within-sector fit, such as mining, utilities, and construction, represent a small share of NOW recipients and are characterised by fundamentally different revenue recognition and cost structures. The overall pattern suggests TWV and Revenue loss are similar for most sectors.

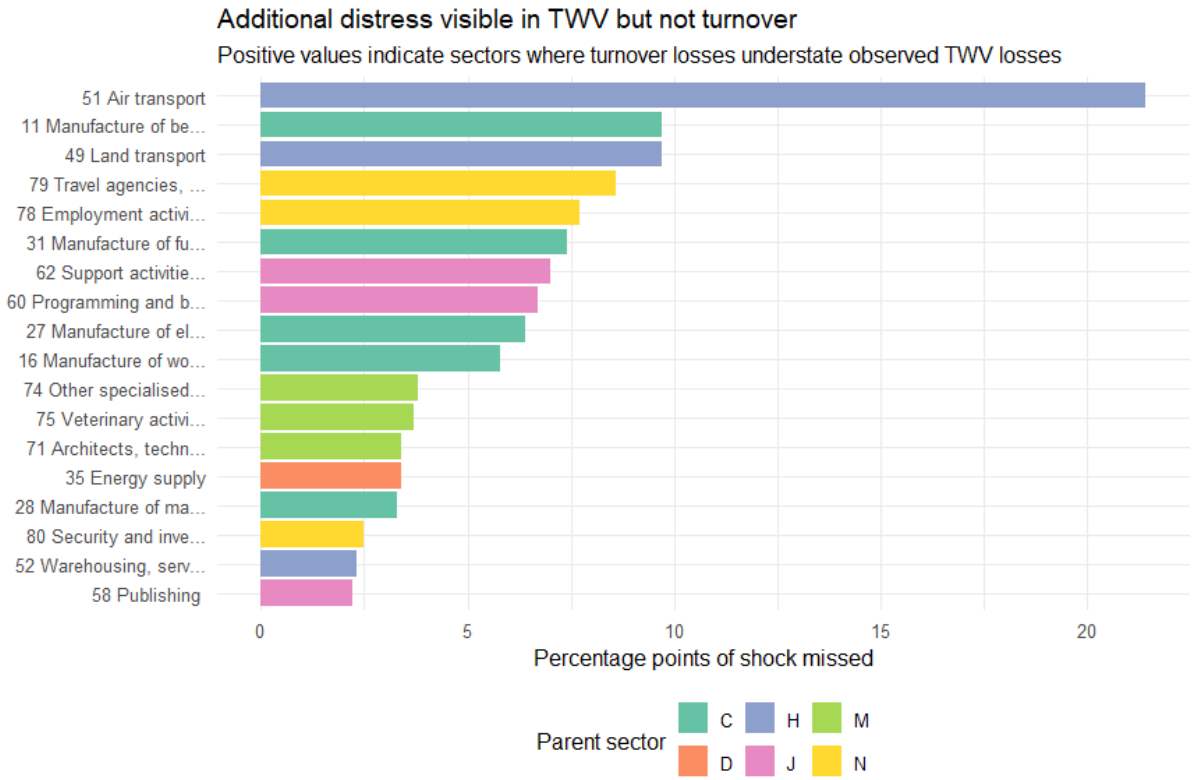


***Figure A2**. Sectors where turnover change understates the TWV shock, measured as the gap in percentage points between observed TWV decline and turnover decline in 2020; air transport shows the largest discrepancy at roughly 21 points, with most other sectors below 10.*

Where does it diverge?

Although turnover change and value added change are strongly correlated overall, Figure A2 shows several sectors where turnover losses understate observed TWV losses. When firm-reported revenue losses are substituted for TWV during deployment, these sectors therefore receive a weaker estimated crisis shock than implied by the underlying change in value added. This may cause genuinely distressed firms to be assigned lower predicted crisis risk.

The divergence is not random but appears to reflect differences in sector cost structures. Because value added equals turnover minus intermediate inputs, the two measures move together when input costs fall alongside sales. Where a larger share of costs remains fixed in the short run, value added declines more rapidly than turnover, causing turnover losses to understate the severity of the shock.

Figure A2 illustrates this pattern. The largest discrepancy occurs in air transport (SBI 51), where turnover declined by 54.1% while value added declined by 75.5%. Smaller but similar differences appear in transport, travel agencies, temporary employment services, and several manufacturing subsectors. These sectors combine relatively high fixed costs with strong exposure to COVID-19 demand shocks.

The divergence is concentrated in a limited number of sectors rather than being widespread across the economy. For most sectors, turnover and value added move closely together, consistent with the strong relationship observed in Figure A1. The results therefore suggest that turnover is a reasonable deployment variable overall, while highlighting sectors where crisis severity may be understated.

Extension

The present analysis uses a single cross-section of 2020 sector-level data. Future work could strengthen this comparison by constructing a panel using multiple years of quarterly CBS observations, providing greater variation for estimating the relationship between turnover and value added across sectors and over time.

Appendix B: Model Specification - Interaction Term Justification

This appendix explains the rationale for the selected interactions and why additional interactions were not included.

Both the exit and employment-loss models include interaction terms only where there is a plausible reason to expect that a firm characteristic modifies the effect of a crisis on employment risk. Limiting interactions to a small set of theoretically motivated variables keeps the models simple, reduces the risk of overfitting, and is intended to improve their ability to generalise to the much larger COVID-19 revenue losses observed at deployment.

Firm size interacts with the crisis dummy because larger firms may be better able to absorb temporary revenue losses through greater financial reserves, better access to credit, or more diversified sources of income, suggesting that the effect of a crisis may differ by firm size.

Firm age interacts with the crisis dummy because older firms may be more resilient to adverse shocks, having accumulated experience, established customer relationships, or financial buffers that could influence how they respond to a crisis.

Full-time share interacts with the crisis dummy because firms with a larger share of permanent employees may have less flexibility to adjust labour costs during a downturn, potentially altering the relationship between a crisis shock and employment loss.

Sector interacts with the crisis dummy because the economic impact of a recession is expected to differ across industries. Interactions are estimated at the one-digit sector level rather than the two-digit SBI level to ensure sufficient observations within each sector.

Female workforce share is not interacted with the crisis dummy because I do not identify a clear theoretical reason why it would systematically modify firms' sensitivity to economic shocks beyond the structural effects already captured in the model.

Revenue loss interactions

Revenue loss enters both models only as a continuous main effect. No interaction terms are included because COVID-19 revenue losses are much larger than those observed during the historical crises used for estimation, making interaction effects difficult to estimate reliably. In addition, firm characteristics such as size and age may already be partly reflected in observed revenue losses, so further interactions may capture the same information twice rather than improving prediction.

B.2 Validating the Sector-Level Shock Measure

This appendix evaluates whether demeaned sector-level value-added (TWV) provides a suitable proxy for crisis intensity and whether the estimated crisis effect is robust to alternative definitions of a crisis year.

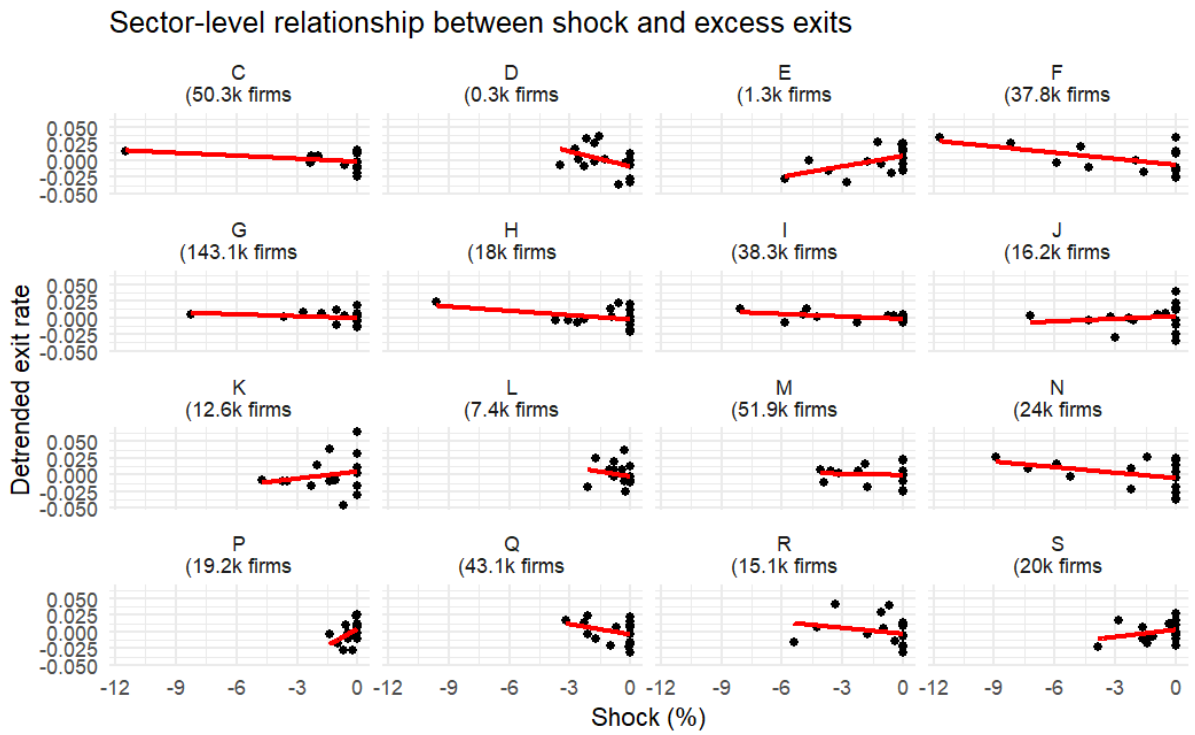


Figure B1. *Sector-level relationship between the estimated crisis shock (x-axis) and detrended firm exit rate (y-axis). Each panel represents one sector, with the number of firms shown beneath the sector code. Black points denote annual sector-level observations, and the red line shows the fitted linear relationship used to assess whether larger sectoral shocks are associated with higher excess firm exits.*

Figure B1 provides the motivation for using demeaned sector-level value-added shocks (TWV) as a proxy for crisis intensity. Across most sectors, larger negative shocks are associated with higher excess exit rates, consistent with the assumption that below-trend sectoral performance captures adverse economic conditions faced by firms.

The relationship is evaluated using the nine largest market sectors (C, F, G, H, I, M, N, R and S), which account for most firms and employment in the Dutch market economy while excluding small or atypical sectors that could disproportionately influence the estimated crisis effect.

Sector	Corr. (threshold −5%)	Obs. (−5%)	Corr. (threshold 0%)	Obs. (0%)
C	-0.300	1	-0.343	7
F	-0.507	4	-0.526	7
G	-0.117	1	-0.221	7
H	-0.463	1	-0.393	8
I	-0.283	3	-0.510	8
M	-	0	-0.069	7
N	-0.360	4	-0.380	7
R	+0.187	1	-0.225	9
S	-	0	+0.288	9
Shock coef.	**−0.00186 (p = 0.001)**		**−0.00197 (p < 0.001)**	

Table B2. *Robustness of sector-level shock coefficient to crisis threshold choice. Corr. reports the within-sector correlation between the demeaned TWV shock and detrended exit rate. Obs. counts sector-year observations qualifying as crisis years under each threshold (−5%: shock below −5%; 0%: shock below 0%). Sectors M and S have no qualifying observations at the −5% threshold (shown as -). The pooled shock coefficient is stable across thresholds, changing from −0.00186 (p = 0.001) to −0.00197 (p < 0.001). The −5% threshold is retained in the final model. Only sectors included in the regression (C, F, G, H, I, M, N, R, S) are shown.*

Table B2 evaluates whether the estimated crisis effect depends on the choice of crisis threshold used to identify downturn years. Relaxing the threshold from −5% to 0% changes the estimated shock coefficient only from −0.00186 to −0.00197, indicating that the relationship between sector-level shocks and firm exits is largely unaffected by the threshold choice. The −5% threshold is therefore retained in the final specification.

Appendix C Revenue loss reconstruction

Firm level revenue loss is not directly observed in the dataset, to reconstruct it is derived from the NOW advance cost of each firm. The now advance cost of each firm is: NOW_cost = revenue_loss × wage_bill x 3 months × subsidy_rate x 0.8.(UWV).
where the subsidy rate is 0.9, the factor 0.8 represents that 80% of the cost is paid upfront, and the wage bill is approximated using sector wages, number of jobs and the duration of 1 round being 3 months.

Rearranging the formula gives: revenue_loss = NOW_cost / (sector_wage x 3 months × subsidy_rate x 0.8).

NOW Round	Number of Firms	Mean Rev. Loss (%)	Median Rev. Loss (%)	% at Cap (100%)
1	51,044	50.0	41.1	20.3
2	26,365	50.7	40.8	21.5
3	29,974	42.8	30.3	16.7
4	29,529	47.3	36.4	19.7
5	18,186	45.4	33.7	18.9
6	11,286	38.0	25.7	13.4
7	15,586	36.8	24.1	13.3
8	12,170	45.7	34.1	19.0

Table C1. *Revenue loss distribution by NOW round. Sample restricted to firms matched to LISA employment register. Cap at 100% refers to firms reporting complete revenue loss.*

Graph C1 shows the mean implied revenue loss is 50% for round 1, consistent with the UWV reported numbers of round 1 and 2. Values exceeding 100% are capped, as declared revenue loss cannot exceed 100% by definition.

Size Category	% at Cap (100%)	% Below 20%	Avg. NOW Budget (EUR)	Number of Firms
1	52.0	14.0	32,010	7,569
2	46.4	13.3	22,201	23,347
3	18.8	32.2	25,661	56,879
4	11.7	32.4	33,584	47,796
5	9.6	29.6	53,351	30,625
6	8.2	29.9	107,628	19,807
7	7.1	35.3	247,064	5,247
8	6.7	39.0	436,533	1,927
9	10.5	38.1	1,162,909	734
10	9.6	39.4	15,171,576	104
11	5.6	22.2	2,807,043	18
12	4.6	56.3	6,310,065	87

Table C2. *Revenue loss reconstruction quality and average NOW budget by CBS establishment size category. The share of firms at the 100% cap is strongly concentrated in the two smallest size categories, while the share below the 20% eligibility threshold is relatively higher for larger firms. Both patterns are consistent with reconstruction error arising from the use of sector-average wages as a proxy for firm-level wage bills. Average NOW budget increases substantially with firm size, reaching several million euros for the largest categories. Categories 10-12 contain few observations and should be interpreted with caution.*

Figure C2 illustrates the distribution of reconstructed revenue losses by firm size. Approximately 20% of firms reach the 100% cap, while around 29% fall below the 20% NOW eligibility threshold. Since all firms in the sample received NOW support, observations below 20% reflect reconstruction error rather than true ineligibility. The error arises because sector-average wages are only an approximation of firm-specific wage bills, causing estimated revenue losses to be either over- or understated. As Table C2 shows, capped observations are concentrated among the smallest firms, whereas firms reconstructed below the 20% threshold are more common among larger firms, indicating that reconstruction error differs systematically across the size distribution.

The reconstructed revenue-loss measure should therefore be interpreted as a noisy proxy rather than an exact estimate of firm-specific revenue decline. In addition to the wage approximation, firms may also have differed in how accurately they reported expected revenue losses when applying for NOW support.

Despite this measurement error, the analysis focuses on aggregate patterns and relative firm rankings rather than exact firm-level revenue losses. Across groups of firms, random over- and underestimation are expected to offset each other, reducing the influence of firm-specific wage differences on average crisis exposure. The main exception is the 100% cap, which is concentrated among the smallest firms and therefore may lead Tier 5 to overstate the concentration of crisis-induced employment loss. Its exact values should therefore be interpreted with additional caution.

Appendix D: Horizon Choice

Choice of prediction horizons

The exit and employment-loss models use different prediction horizons because the underlying adjustment processes differ. This appendix explains the rationale for using a three-year horizon for the exit model and a one-year horizon for the employment-loss model.

Exit model: three-year horizon

The exit model uses a three-year outcome horizon for two reasons.

First, firms facing the same revenue shock are expected to exit at different times depending on their financial resilience. Some may fail after 12 months, while others survive for several years before exiting. A one-year horizon would therefore classify many firms that ultimately fail as unaffected by the shock, whereas a three-year horizon captures a broader range of crisis-induced exits.

Employment-loss model: one-year horizon

The employment-loss model uses a one-year outcome horizon because employment adjustment is expected to occur more rapidly than firm exit. Firms can reduce employment through layoffs, attrition or contract adjustments shortly after the onset of a shock. Extending the horizon would increasingly mix the immediate crisis response with subsequent recovery or longer-term structural changes, making it more difficult to isolate the employment adjustment associated with the initial crisis.

Appendix E: Why Round 1 Is Used for the Crisis-Induced Employment-Loss Analysis

This appendix explains why Round 1 NOW applications are used to estimate crisis-induced employment loss in Sections 5.1–5.3 rather than combining revenue losses across all eight subsidy rounds.

Why Round 1?

The framework estimates crisis-induced employment loss using firms' reported revenue losses. Round 1 revenue losses largely reflect the initial economic shock before the continued effects of the NOW programme could materially influence firms' financial position. In later rounds, however, reported revenue losses are no longer independent of the intervention itself. By supporting firms' liquidity and employment, NOW likely reduced subsequent revenue losses relative to the no-intervention scenario.

Because reported revenue loss is an input to the crisis component of the model, using later-round revenue losses would partially incorporate the effects of the programme into the estimated crisis shock, thereby understating the employment loss that might have occurred without support.

Implications

Round 1 therefore provides the closest available approximation to the initial untreated crisis shock and is used as the basis for estimating crisis-induced employment loss and the associated expected WW costs in Sections 5.1–5.3.

Using all eight NOW rounds remains a feasible alternative under the assumption of a longer crisis duration. However, doing so requires assumptions about the counterfactual evolution of firms' revenue losses after Round 1, since the observed revenue losses in later rounds are already affected by NOW support. The study therefore limits the counterfactual analysis to the approximately three-month duration of Round 1, while using information from all eight rounds separately in Section 5.4 to evaluate alternative firm-ranking methods.

Direction of Bias

Both approaches are conservative. Restricting the analysis to Round 1 excludes additional crisis exposure that occurred later in the pandemic, while using observed revenue losses from later rounds incorporates the effects of the programme itself, reducing the measured crisis component. The true counterfactual crisis-induced employment loss in the absence of intervention is therefore likely to exceed both estimates.

Appendix F. Employment Loss Model: Specification and Validation

The employment loss model predicts the probability that a surviving firm reduces its headcount in the following year. This appendix explains the choice of outcome variable, the reasons for modelling the probability of shrinkage rather than the direct magnitude of employment loss, the sample restriction to firms with three or more employees, the estimation of the jobs-lost multiplier, and the validation of the resulting job loss estimates.

F.1 Why probability of shrinkage rather than magnitude

Employment loss among surviving firms is highly skewed. Most firms experience little or no employment reduction in a given year, while a small number account for large contractions. Directly modelling the magnitude of employment change therefore produces unstable predictions, including values that can fall below zero or exceed firm size.

A more fundamental problem emerged when revenue loss was added to the magnitude regression: the rev_loss coefficient showed the wrong sign, with larger revenue losses associated with less employment reduction rather than more. This may reflect survivorship bias. The regression sample includes only firms that survived, and firms surviving severe shocks may be more likely to recover employment in subsequent years, creating a positive relationship between shock severity and observed employment change.

In the logit specification predicting the probability of shrinkage, the revenue-loss coefficient is negative and correctly signed (−0.014), indicating that larger shocks increase employment risk as expected. Expected employment loss is then calculated as the predicted probability of shrinkage multiplied by the average number of jobs lost conditional on shrinkage. Revenue loss therefore affects aggregate employment loss through the probability term: larger revenue losses increase the probability of shrinkage and, consequently, the aggregate number of expected jobs lost. Whether this relationship continues to scale appropriately for the substantially larger COVID-19 revenue losses observed during programme implementation cannot be verified directly and therefore remains an assumption of the framework. Table F3 summarises the trade-offs across the alternative approaches considered.

F.2 Sample restriction to firms with three or more employees

The training sample is restricted to firms with three or more employees. Firms with exactly two employees are excluded because a two-employee firm that reduces headcount falls to one employee. Since firms with one employee are excluded from the panel throughout, this outcome is never observed in the data: the firm simply disappears from the employment register, making shrinkage indistinguishable from exit. Including two-employee firms as training observations would therefore allow the model to partially learn exit patterns through the shrinkage channel.

Firms with three employees can reduce headcount to two, which remains observable in the panel. In principle, a separate two-employee category could be modelled, but these firms represent a small share of employment and excluding them likely has a negligible effect on the aggregate results. At deployment, firms with two or fewer employees are assigned a

shrinkage probability of zero, so their employment risk is captured exclusively through the exit channel.

F.3 Jobs lost multiplier

Expected jobs lost per shrinking firm is estimated directly from the training data as the average actual jobs lost among firms that shrank, grouped by exact firm size up to 19 employees. Above 19 employees, thin samples produce non-monotonic estimates, so a single pooled average is used for all firms with 20 or more employees. Table F1 reports the resulting multipliers.

jobs_group	avg_jobs_lost	Note
3	1.000	Minimum: can only lose 1 job
4	1.295	
5	1.449	
6	1.608	
7	1.743	
8	1.894	
9	2.025	
10	2.210	
11	2.275	
12	2.433	
13	2.500	
14	2.656	
15	2.809	
16	2.823	
17	2.974	
18	3.089	
19	3.190	Monotonic, individual rates used
20+	10.543	Non-monotonic above 19, pooled avg used

Table F1. *Average jobs lost per shrinking firm by firm size group. Individual rates are used for firms with 3–19 employees where the relationship is monotonically increasing. A pooled average of 10.54 is used for firms with 20 or more employees where thin samples produce non-monotonic estimates.*

F.4 Aggregate calibration

Table F2 validates the combined estimate (predicted shrinkage probability multiplied by the size-specific jobs lost multiplier) against actual observed job losses in the training data. Firms are ranked into deciles by predicted jobs lost and mean predicted and actual job loss are compared within each decile.

The ranking is monotonically correct across all ten deciles, confirming the model correctly orders firms by expected employment loss. At the aggregate level, total predicted jobs lost multiplied by 4.65 (the population average jobs lost per shrinking firm) equals 10.909, compared with an actual total of 10.816, a difference of less than 1 percent. This confirms near-perfect aggregate calibration for the training data. The multiplier approach is therefore used exclusively for aggregate expected job destruction in the 5 tiers, rather than firm-level point predictions.

Bucket	Mean pred	Mean actual	Pred × 4.65	Actual × 4.65
1	0.118	0.099	0.549	0.460
2	0.146	0.121	0.679	0.563
3	0.163	0.142	0.758	0.660
4	0.180	0.171	0.837	0.795
5	0.198	0.198	0.921	0.921
6	0.219	0.235	1.018	1.093
7	0.244	0.274	1.135	1.274
8	0.278	0.318	1.293	1.479
9	0.332	0.345	1.544	1.604
10	0.468	0.423	2.176	1.967
Total	**2.346**	**2.326**	**10.909**	**10.816**

Table F2. *Predicted vs actual shrinkage probability by decile, with implied jobs lost scaled by 4.65 (population average jobs lost per shrinking firm). Total predicted × 4.65 = 10.909 vs actual × 4.65 = 10.816, a difference of 0.9%, confirming near-perfect aggregate calibration at the population level.*

F.5 Summary of approaches considered

Table F3 summarises the trade-offs across the four approaches considered.

Approach	Benefit	Downside
OLS continuous regression	Direct magnitude prediction	Survivorship bias flips rev_loss sign; extreme residuals make fit poor
GLM probability only	Correct signs, AUC 0.65, stable coefficients	No job loss magnitude, needs second step
GLM × flat 4.65 avg	Aggregate ratio 1.003, simple	Biased when firm size distribution shifts between training and deployment
GLM × exact jobs_group multiplier (3–19 individual, 20+ pooled)	Aggregate ratio 1.003, size-corrected, monotonic up to 19 jobs	Non-monotonic above 20 jobs due to thin samples, pooled there

Table F3. *Employment loss magnitude approach comparison. The GLM probability model combined with a size-specific jobs lost multiplier is the preferred approach, producing an aggregate calibration ratio of 1.003 while correctly identifying the direction of crisis-induced shrinkage risk.*

Appendix G. Exit Model: Coefficients and Stability

This appendix reports the full exit-model coefficients and examines the stability of the estimated relationships under alternative crisis specifications. The stability analysis compares models with and without reported revenue loss while retaining the crisis dummy to evaluate whether both variables capture similar information and to justify the final model specification.

Variable	*Estimate*	*Std. Error*	*z value*	*Pr(>|z|)*	*Direction*
Firm characteristics					
(Intercept)	*-0.438*	*0.0078*	*-56.32*	*<2e-16 ****	—
log(number_of_jobs)	*-0.458*	*0.0011*	*-427.24*	*<2e-16 ****	*↑ Smaller firms more likely to exit*
age_bucket 0–3	*0.667*	*0.0035*	*191.27*	*<2e-16 ****	*↑ Young firms more likely to exit*
age_bucket 4–7	*0.253*	*0.0035*	*72.48*	*<2e-16 ****	↑
age_bucket 8–15	*0.065*	*0.0035*	*18.48*	*<2e-16 ****	↑
ft_share	*-0.353*	*0.0052*	*-67.32*	*<2e-16 ****	*↓ Higher full-time share reduces exit risk*
female_share	*-0.232*	*0.0072*	*-32.51*	*<2e-16 ****	*↓ Higher female share reduces exit risk*
female_ft_share	*0.093*	*0.0080*	*11.59*	*<2e-16 ****	↑
Crisis variable					
crisis_dummy	—	—	—	—	*Not included (collinear with rev_loss; see Table G2)*
rev_loss_market	*-0.016*	*0.0004*	*-36.53*	*<2e-16 ****	*↑ Larger revenue loss increases exit risk*
Sector fixed effects (ref: sector C)					
D -Energy supply	*0.857*	*0.0360*	*23.82*	*<2e-16 ****	↑
E -Water supply & waste management	*0.309*	*0.0187*	*16.56*	*<2e-16 ****	↑
F -Construction	*0.309*	*0.0044*	*70.69*	*<2e-16 ****	↑
G -Wholesale & retail trade	*0.327*	*0.0034*	*95.85*	*<2e-16 ****	↑
H -Transportation & storage	*0.499*	*0.0054*	*93.00*	*<2e-16 ****	↑
I -Accommodation & food service	*0.225*	*0.0044*	*50.83*	*<2e-16 ****	↑
J -Information & communication	*0.713*	*0.0052*	*137.67*	*<2e-16 ****	↑
K -Financial & insurance services	*0.691*	*0.0058*	*118.19*	*<2e-16 ****	↑

L -Real estate activities	*0.339*	*0.0075*	*45.05*	*<2e-16 ****	↑
M -Professional, scientific & technical	*0.516*	*0.0039*	*132.37*	*<2e-16 ****	↑
N -Administrative & support services	*0.583*	*0.0047*	*122.85*	*<2e-16 ****	↑
O -Public administration & defence	*0.755*	*0.0126*	*59.93*	*<2e-16 ****	↑
P -Education	*0.272*	*0.0059*	*46.18*	*<2e-16 ****	↑
Q -Human health & social work	*0.079*	*0.0047*	*16.83*	*<2e-16 ****	↑
R -Arts, recreation & entertainment	*0.283*	*0.0057*	*49.75*	*<2e-16 ****	↑
S -Other service activities	*0.226*	*0.0054*	*42.21*	*<2e-16 ****	↑
Province fixed effects (ref: Groningen)					
Friesland (Fryslân)	*-0.741*	*0.0062*	*-119.30*	*<2e-16 ****	↓ *Reduces exit risk vs reference province*
Drenthe	*-0.474*	*0.0064*	*-73.60*	*<2e-16 ****	↓
Overijssel	*-0.650*	*0.0055*	*-117.52*	*<2e-16 ****	↓
Flevoland	*-0.257*	*0.0070*	*-36.82*	*<2e-16 ****	↓
Gelderland	*-0.402*	*0.0049*	*-82.18*	*<2e-16 ****	↓
Utrecht	*-0.437*	*0.0053*	*-82.40*	*<2e-16 ****	↓
Noord-Holland	*-0.278*	*0.0046*	*-60.69*	*<2e-16 ****	↓
Zuid-Holland	*-0.566*	*0.0047*	*-120.52*	*<2e-16 ****	↓
Zeeland	*-0.503*	*0.0069*	*-73.28*	*<2e-16 ****	↓
Noord-Brabant	*-0.392*	*0.0047*	*-82.50*	*<2e-16 ****	↓
Limburg	*-0.627*	*0.0056*	*-112.57*	*<2e-16 ****	↓

Model fit: null deviance 8,888,136 on 8,531,861 df; residual deviance 8,369,337 on 8,531,826 df; AIC 8,369,409; n = 8,531,862.

Table G1. *Exit model GLM coefficients with direction of effect on firm exit probability. Sector fixed effects are relative to sector C -Manufacturing; all reported sectors increase exit risk relative to this baseline. Province fixed effects are relative to Groningen; all reported provinces reduce exit risk relative to the reference. The crisis dummy is excluded from the final exit model due to collinearity with revenue loss, and is retained only in the stability check (Table G2). Revenue loss enters as a negative number, so its negative coefficient implies that larger revenue losses increase exit risk. All reported coefficients significant at p < 0.001 (***). Reference category for age is 16+ years. Estimated on firms with 2 or more employees, training years 2000–2016.*

Coefficient Stability

Variable	Est. (base)	Est. (+ rev_loss)	% change	Sig. base	Sig. full
Firm characteristics					
(Intercept)	-0.395	-0.395	0.2%	*	*
log(number_of_jobs)	-0.457	-0.457	0.0%	*	*
age_bucket 0–3	0.677	0.676	-0.1%	*	*
age_bucket 4–7	0.238	0.237	-0.3%	*	*
age_bucket 8–15	0.065	0.065	0.0%	*	*
ft_share	-0.365	-0.365	0.0%	*	*
female_share	-0.228	-0.229	-0.1%	*	*
female_ft_share	0.088	0.089	0.2%	*	*
Crisis variables					
crisis_dummy	-0.212	-0.214	-0.8%	*	*
rev_loss_market	-	-0.005	-	-	*
Crisis dummy interactions					
age_bucket 0–3 × crisis	0.030	0.026	-14.4%	*	*
age_bucket 4–7 × crisis	0.225	0.219	-2.4%	*	*
age_bucket 8–15 × crisis	0.039	0.031	-20.7%	*	*
log(number_of_jobs) × crisis	0.003	0.003	-13.7%		
ft_share × crisis	0.067	0.067	-0.6%	*	*

Table G2. Stability of exit-model coefficients under alternative crisis specifications. *Est. (base)* reports the model including the crisis dummy and its interactions; *Est. (+ rev_loss)* additionally includes reported revenue loss. Structural firm-characteristic coefficients remain virtually unchanged after adding revenue loss, while the crisis interactions change only modestly and retain the same substantive interpretation. In contrast, the revenue-loss coefficient is small (-0.005) when the crisis dummy is included, suggesting that both variables capture similar crisis-period information. Although the crisis-dummy specification achieves a slightly lower training AIC (8,363,290 versus 8,369,409), the crisis dummy identifies the specific COVID-19 period and therefore cannot be used when extrapolating the model to future crises. The final exit model consequently omits the crisis dummy and retains reported revenue loss as the extrapolatable measure of crisis exposure. The log(number_of_jobs) × crisis interaction is not statistically significant; all other reported coefficients are significant at $p < .05$ (*). Sector and province fixed effects are omitted.

Appendix H Shrinkage Model Coefficients and Stability

This appendix reports the full employment-shrinkage model coefficients and examines the stability of the estimated relationships under alternative crisis specifications. The stability analysis compares models with and without reported revenue loss while retaining the crisis dummy to assess whether reported revenue loss contributes additional information beyond the common crisis effect and to justify the final model specification.

Variable	*Estimate*	*Std. Error*	*z value*	*Pr(>|z|)*	*Direction*
Firm characteristics					
(Intercept)	*-1.798*	*0.0136*	*-132.442*	*<2e-16 ****	*-*
log(number_of_jobs)	*0.437*	*0.0013*	*342.067*	*<2e-16 ****	*↑ Larger firms more likely to shrink*
age_bucket 0–3	*-0.120*	*0.0060*	*-19.808*	*<2e-16 ****	*↓ Young firms less likely to shrink*
age_bucket 4–7	*0.093*	*0.0054*	*17.114*	*<2e-16 ****	*↑*
age_bucket 8–15	*0.124*	*0.0054*	*23.096*	*<2e-16 ****	*↑*
ft_share	*-0.301*	*0.0100*	*-30.009*	*<2e-16 ****	*↓ Higher full-time share reduces shrink risk*
female_share	*0.111*	*0.0122*	*9.100*	*<2e-16 ****	*↑ Higher female share increases shrink risk*
female_ft_share	*-0.101*	*0.0137*	*-7.411*	*1.25e-13 ****	*↓*
Crisis variables					
crisis_dummy	*-0.053*	*0.0167*	*-3.202*	*0.001367 ***	*↑ Crisis increases shrink risk (negative coef on logit scale)*
rev_loss_market	*-0.014*	*0.0007*	*-19.637*	*<2e-16 ****	*↑ Larger revenue loss increases shrink risk*
Sector fixed effects (ref: sector C Manufacturing)					
D - Energy supply	*-0.268*	*0.0510*	*-5.248*	*1.54e-07 ****	*↓*
E - Water supply & waste management	*-0.060*	*0.0225*	*-2.647*	*0.008128 ***	*↓*
F - Construction	*0.124*	*0.0064*	*19.500*	*<2e-16 ****	*↑*
G - Wholesale & retail trade	*-0.059*	*0.0051*	*-11.608*	*<2e-16 ****	*↓*
H - Transportation & storage	*-0.088*	*0.0082*	*-10.715*	*<2e-16 ****	*↓*
I - Accommodation & food service	*-0.140*	*0.0067*	*-20.748*	*<2e-16 ****	*↓*
J - Information & communication	*-0.053*	*0.0100*	*-5.277*	*1.31e-07 ****	*↓*
K - Financial & insurance services	*0.079*	*0.0096*	*8.241*	*<2e-16 ****	*↑*
L - Real estate activities	*-0.015*	*0.0126*	*-1.156*	*0.247801*	*↓ (n.s.)*

M - Professional, scientific & technical	*0.027*	*0.0065*	*4.184*	*2.86e-05 ****	↑
N - Administrative & support services	*-0.075*	*0.0079*	*-9.499*	*<2e-16 ****	↓
O - Public administration & defence	*-0.429*	*0.0131*	*-32.714*	*<2e-16 ****	↓
P - Education	*-0.094*	*0.0074*	*-12.628*	*<2e-16 ****	↓
Q - Human health & social work	*-0.264*	*0.0066*	*-39.736*	*<2e-16 ****	↓
R - Arts, recreation & entertainment	*-0.067*	*0.0100*	*-6.714*	*1.90e-11 ****	↓
S - Other service activities	*0.030*	*0.0088*	*3.361*	*0.000778 ****	↑
Crisis dummy interactions					
sector D × crisis	*0.081*	*0.0957*	*0.850*	*0.395503*	↑ *(n.s.)*
sector E × crisis	*-0.167*	*0.0450*	*-3.702*	*0.000214 ****	↓
sector F × crisis	*-0.019*	*0.0129*	*-1.499*	*0.133951*	↓ *(n.s.)*
sector G × crisis	*-0.036*	*0.0096*	*-3.787*	*0.000152 ****	↓
sector H × crisis	*-0.057*	*0.0163*	*-3.476*	*0.000508 ****	↓
sector I × crisis	*-0.070*	*0.0130*	*-5.341*	*9.23e-08 ****	↓
sector J × crisis	*-0.009*	*0.0190*	*-0.455*	*0.648774*	↓ *(n.s.)*
sector K × crisis	*-0.102*	*0.0198*	*-5.148*	*2.63e-07 ****	↓
sector L × crisis	*0.120*	*0.0246*	*4.850*	*1.23e-06 ****	↑
sector M × crisis	*0.118*	*0.0122*	*9.661*	*<2e-16 ****	↑
sector N × crisis	*0.058*	*0.0150*	*3.846*	*0.000120 ****	↑
sector O × crisis	*0.061*	*0.0264*	*2.325*	*0.020086 **	↑
sector P × crisis	*0.101*	*0.0139*	*7.282*	*3.30e-13 ****	↑
sector Q × crisis	*0.082*	*0.0117*	*6.986*	*2.82e-12 ****	↑
sector R × crisis	*0.075*	*0.0189*	*3.942*	*8.07e-05 ****	↑
sector S × crisis	*-0.013*	*0.0169*	*-0.749*	*0.454104*	↓ *(n.s.)*
age_bucket 0–3 × crisis	*-0.187*	*0.0097*	*-19.255*	*<2e-16 ****	↓ *Young firms face less additional shrink risk in crisis*
age_bucket 4–7 × crisis	*-0.322*	*0.0098*	*-32.731*	*<2e-16 ****	↓
age_bucket 8–15 × crisis	*-0.238*	*0.0080*	*-29.779*	*<2e-16 ****	↓
ft_share × crisis	*0.218*	*0.0121*	*18.086*	*<2e-16 ****	↑ *Higher full-time share increases shrink risk in crisis*

log(number_of_jobs) × crisis	*0.060*	*0.0025*	*23.793*	*<2e-16 ****	*↑ Larger firms face more shrink risk in crisis*
Province fixed effects (ref: Groningen)					
Friesland (Fryslân)	*-0.166*	*0.0089*	*-18.531*	*<2e-16 ****	↓
Drenthe	*-0.220*	*0.0096*	*-23.019*	*<2e-16 ****	↓
Overijssel	*-0.050*	*0.0080*	*-6.287*	*3.23e-10 ****	↓
Flevoland	*0.206*	*0.0101*	*20.420*	*<2e-16 ****	↑
Gelderland	*0.111*	*0.0074*	*14.903*	*<2e-16 ****	↑
Utrecht	*-0.085*	*0.0079*	*-10.797*	*<2e-16 ****	↓
Noord-Holland	*-0.495*	*0.0075*	*-66.081*	*<2e-16 ****	↓
Zuid-Holland	*-0.211*	*0.0073*	*-29.042*	*<2e-16 ****	↓
Zeeland	*-0.163*	*0.0099*	*-16.506*	*<2e-16 ****	↓
Noord-Brabant	*-0.238*	*0.0074*	*-32.226*	*<2e-16 ****	↓
Limburg	*-0.209*	*0.0081*	*-25.911*	*<2e-16 ****	↓

Model fit: null deviance 4,769,755 on 4,376,663 df; residual deviance 4,530,503 on 4,376,606 df; AIC 4,530,619; n = 4,376,664.

Table H1. *Employment loss model GLM coefficients with direction of effect on shrinkage probability. Sector fixed effects are relative to sector C - Manufacturing; province fixed effects are relative to Groningen. Crisis dummy interactions allow the crisis effect to vary by sector, firm age, size, and full-time share. Revenue loss enters as a negative number, so the negative coefficient (−0.014) implies larger revenue losses increase shrinkage risk. Significance codes: *** p<0.001, ** p<0.01, * p<0.05; coefficients marked (n.s.) are not significant at p<0.05.*

Coefficient Stability

Variable	Est. (base)	Est. (+ rev_loss)	% change	Sig. base	Sig. full
Firm characteristics					
(Intercept)	-2.627	-2.625	0.1%	*	*
log(number_of_jobs)	0.673	0.672	0.0%	*	*
age_bucket 0–3	-0.188	-0.190	-1.1%	*	*
age_bucket 4–7	0.079	0.076	-3.2%	*	*
age_bucket 8–15	0.126	0.126	-0.1%	*	*
ft_share	-0.251	-0.252	-0.2%	*	*
female_share	0.214	0.213	-0.4%	*	*

female_ft_share	-0.107	-0.106	0.8%	*	*
Crisis variables					
crisis_dummy	-0.055	-0.070	-26.2%	*	*
rev_loss_market	-	-0.014	-	-	*
Crisis dummy interactions					
age_bucket 0–3 × crisis	-0.209	-0.220	-5.3%	*	*
age_bucket 4–7 × crisis	-0.330	-0.347	-5.2%	*	*
age_bucket 8–15 × crisis	-0.219	-0.246	-12.2%	*	*
ft_share × crisis	0.255	0.253	-0.9%	*	*
log(number_of_jobs) × crisis	0.060	0.059	-1.0%	*	*

Table H2. *Stability of employment-loss model coefficients under alternative crisis specifications. Est. (base) reports the model including the crisis dummy and its interactions; Est. (+ rev_loss) additionally includes firm-declared revenue loss. Structural coefficients remain virtually unchanged after adding revenue loss, while the revenue-loss coefficient remains substantial and statistically significant. This indicates that reported revenue loss and the crisis dummy capture complementary information rather than the same crisis signal. Consequently, both variables are retained in the final employment-loss model. Sector and province fixed effects are omitted.*

Appendix I

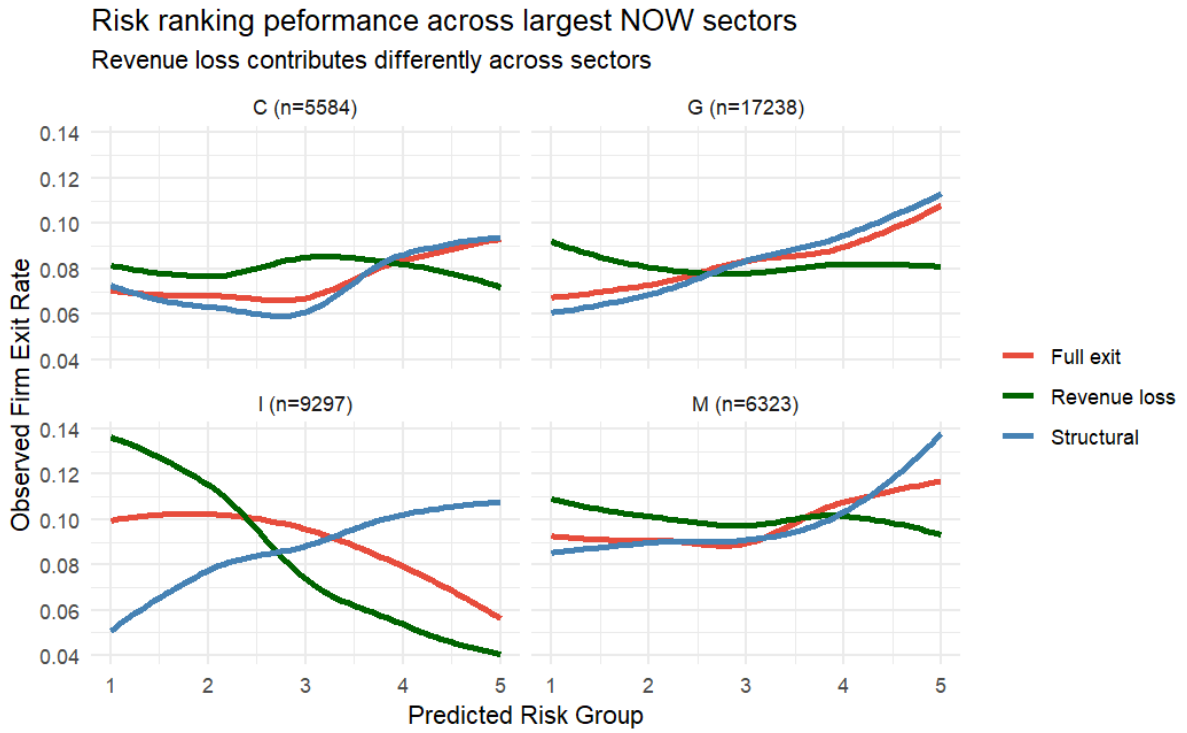


Figure I1. *repeats the comparison from Figure 5.7 separately for the four largest NOW sectors. Within each panel, firms are divided into five predicted-risk groups, and the observed firm exit rate by 2022 is plotted for the structural, revenue-loss and full rankings.*

Figure I1 shows that the aggregate pattern masks important sectoral differences. Structural risk provides the clearest separation between low- and high-risk firms in sectors G and M while maintaining a broadly monotonic relationship with observed exit rates, and performs similarly to, or slightly better than, the full model. In sector C, the structural and full rankings are almost identical, indicating that reported revenue loss contributes little additional discriminatory information. Sector I differs in that both structural risk and revenue loss produce monotonic rankings, but in opposite directions, suggesting that they capture different dimensions of firm vulnerability. Similar revenue-loss patterns also appear in several smaller sectors (Appendix I), although its discriminatory performance is considerably less consistent across industries than that of the structural ranking.

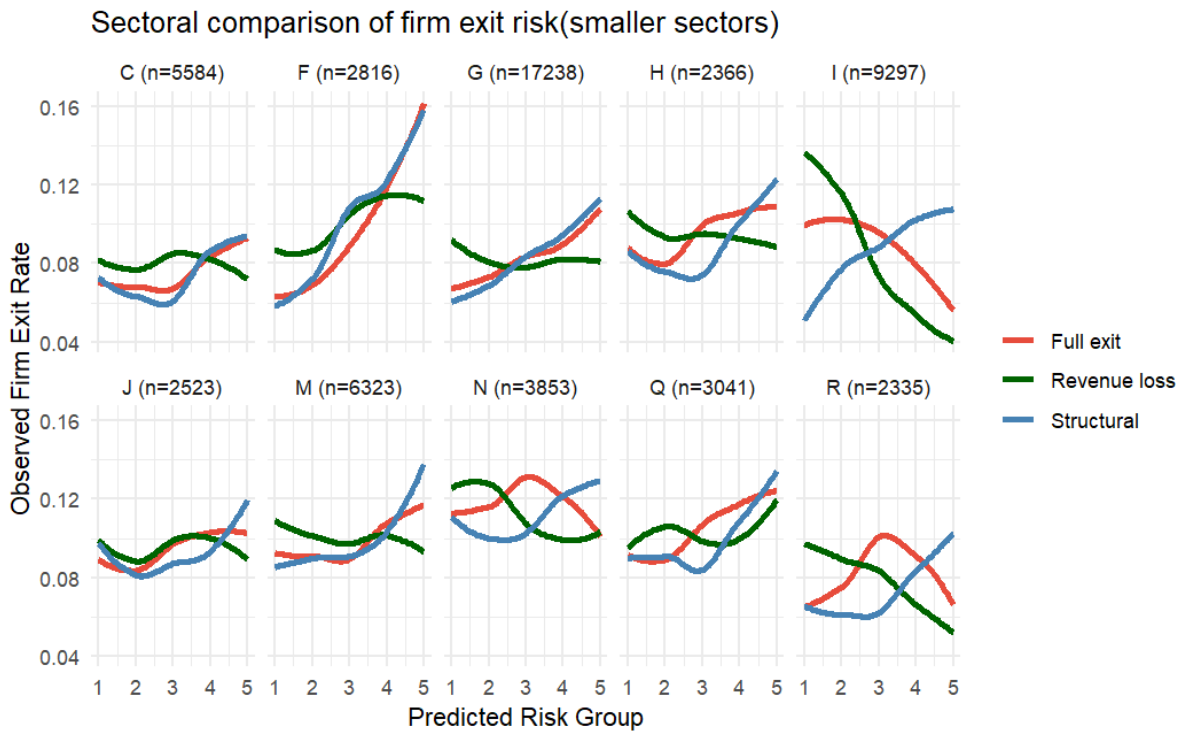


Figure I2. *Observed firm exit rate by 2022 across five predicted risk groups (1 = lowest, 5 = highest) under three rankings (structural, revenue loss, full exit model), shown for the ten largest NOW sectors; structural risk is non-monotonic in every panel, while the rankings using revenue loss rise more consistently.*

Appendix I examines the remaining ten sectors separately. Unlike the aggregate analysis, no consistent pattern emerges across the smaller sectors. The rankings produced by all three models vary considerably between sectors, reflecting the substantially smaller number of observations available within each five-group comparison. Consequently, the appendix is not informative about the relative performance of the three models at the sector level. Rather, it shows that the aggregate ordering is not driven by any single sector but instead reflects patterns that emerge only when information is pooled across firms.

Appendix J

Sector	Description	Wage W_comp (€)	Cap φ	Duration Du (mo.)	Wage measured 2nd digit (codes)	Duration measured 1st digit (codes)
A	Agriculture, forestry & fishing	3,764	0.99	9.2	3/3	Yes
B	Mining & quarrying	10,198	0.48	12.2	0/3	No
C	Manufacturing	5,184	0.91	8.4	24/24	24/24
D	Energy supply	6,857	0.71	12.2	1/1	No
E	Water & waste	5,350	0.92	8.4	4/4	No
F	Construction	5,167	0.92	10.2	3/3	Yes
G	Wholesale & retail	3,362	1.00	9.0	3/3	Yes
H	Transport & storage	4,088	0.99	7.7	5/5	Yes
I	Hospitality	1,283	1.00	7.5	0/2	2/2
J	Information & communication	5,525	0.88	12.2	6/6	No
K	Financial services	7,777	0.63	12.2	3/3	Yes
L	Real estate	4,855	1.00	9.1	1/1	No
M	Professional services	4,812	0.92	9.1	7/7	Yes
N	Administrative services	3,043	1.00	7.1	6/6	Yes
O	Public administration	5,309	0.92	10.4	1/1	Yes
P	Education	4,364	1.00	8.7	1/1	Yes
Q	Health & social work	3,268	1.00	9.1	3/3	Yes
R	Arts & recreation	2,819	1.00	7.7	4/4	Yes
S	Other services	3,183	1.00	9.1	3/3	Yes

Table J1. *Sector-level inputs to the cost calculation (2019-2020). W_comp is the average monthly wage including employer social contributions; φ is the benefit-cap factor (φ < 1 where the sector wage exceeds the maximum dagloon, so benefits apply only up to the ceiling); Du is the expected unemployment duration in months. The final two columns report how many of the sector's constituent two-digit SBI codes use a directly measured value rather than one borrowed from a parent or related sector (shaded where any are borrowed). Wages from CBS National Accounts; durations from UWV records. Values are shown at sector-letter level as unweighted means across two-digit codes; the cost calculation applies the underlying two-digit values, which are provided in the replication data.*

Appendix K. Estimating Sector-Specific Unemployment Duration

For each month a laid-off worker remains unemployed, the government incurs two costs, both expressed as a share of the sector's average monthly labour cost W_comp, measured as gross wage plus employer social contributions.

The first is the unemployment benefit. Dutch law sets WW at 70 percent of the prior wage, giving a benefit term of W_comp × φ × 0.70. The factor φ caps the wage at the maximum daily wage on which benefits are paid (the *maximumdagloon*, approximately €4,700 per month over the study period), since no benefit accrues on the portion of a wage above this ceiling, even though NOW reimbursed employers on wages up to roughly twice this amount. It is defined as φ = min(1, ceiling / W_comp), so it scales the wage down only in sectors whose average wage exceeds the ceiling. Because the statutory 70 percent applies to gross wage while W_comp is measured as total labour cost, this term somewhat overstates the benefit actually paid; the resulting WW-cost estimates are therefore conservative in the upward direction.

The second is the labour-tax revenue no longer collected when a wage is lost, comprising income tax and employee and employer social contributions. This is applied as 0.37 of W_comp, the Netherlands' average tax wedge on labour, which was approximately 37–38 percent over the study period (OECD, 2025). The tax wedge is defined as a share of total labour cost, consistent with the definition of W_comp, so this term and its denominator are aligned.

A single national tax-wedge rate is applied rather than sector-specific wedges. In sectors with many young or part-time workers the effective wedge is likely lower, and in those with older or higher-paid workers higher, but because the objective is aggregate fiscal exposure rather than sector-specific tax receipts, a single average is a reasonable approximation. Combining the two components, each month of unemployment costs the government roughly 1.07 times the monthly wage per worker: 70 percent in benefits and about 37 percent in foregone labour-tax revenue.

Sector-specific unemployment durations are estimated from UWV administrative data (December 2019) using the number of ongoing unemployment-benefit recipients (WW *Lopend*) and the monthly number leaving unemployment (WW *Uitstroom*). Comparable sector-level stock and outflow data are not available for the 2008–2013 crisis period, so the most recent pre-COVID observations are used to approximate unemployment durations under normal labour-market conditions. Average unemployment duration is estimated as the ratio of the unemployment stock to the monthly outflow:

Du = WW Lopend / WW Uitstroom

This provides an estimate of the average time a recipient remains in WW under normal labour-market conditions. The resulting durations are mapped from the UWV sector classification to the CBS sector classification used throughout the analysis. Where no direct mapping is available, the closest related sector is used. The final durations used in the treasury-cost calculations are reported in Appendix J.

CBS sector	WW Lopend	WW Uitstroom	Duration (months)
A	3,986	435	9.16
C	29,149	3,474	8.39
F	3,273	322	10.16
G	34,055	3,798	8.97
H	15,498	2,019	7.68
I	9,484	1,260	7.53
K	9,590	788	12.17
M	46,841	5,162	9.07
N	23,467	3,298	7.12
O	1,878	181	10.38
P	9,353	1,080	8.66
Q	30,071	3,309	9.09
R	2,757	360	7.66
S	3,387	372	9.10

Table K.1. *UWV unemployment durations by sector.*

CBS sector	Duration assigned from
B (Mining & quarrying)	K (Financial services)
D (Energy supply)	K (Financial services)
E (Water & waste)	C (Manufacturing)
J (Information & communication)	K (Financial services)
L (Real estate)	M (Professional services)

Table K.2*. Sector mappings used where no direct UWV duration was available.*

The estimated durations should be interpreted as average observed unemployment durations under normal labour-market conditions rather than structural job-finding times. During severe recessions unemployment durations may increase as job-finding rates decline, while exits from WW reflect both re-employment and benefit exhaustion. The estimates therefore provide a practical approximation for calculating expected WW costs rather than a dynamic model of unemployment duration

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